THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF ENGINEERING & TECHNOLOGY
(RSET)

DEPARTMENT OF MECHANICAL ENGINEERING

COURSE STRUCTURE & SYLLABUS FOR UNDERGRADUATE DEGREE IN MECHANICAL ENGINEERING

FOR

Bachelor of Technology
in
Mechanical Engineering

W.E.F

AY (2026 – 27)

STRUCTURE OF THE SYLLABUS FOR A 4 YEAR UG PROGRAMME FROM AY 2026-27

8

SEMESTER I

Sl.No	Course Title	Course code	L	T	P	Credits
1.	Chemistry	CHY022C101	3	0	0	3
2	Mathematics-I	MAT022C101	3	1	0	4
3	Chemistry Lab	CHY022C111	0	0	2	1
4	Fundamentals of Programming	CEE022C103	3	0	0	3
5	Fundamentals of Programming Lab	CEE022C113	0	0	2	1
6	Engineering Graphics & Design Lab	CEE022C114	0	0	4	2
7	Manufacturing Practice Workshop	MEE022C115	0	0	4	2
8	Universal Human Values	BHS022A106	2	0	0	2
9	Sports and Yoga/NSS/NCC	CEE022S117	0	0	2	1
	Total Credit					19

SEMESTER II

Sl.No	Course Title	Course code	L	T	P	Credits
1.	Physics	PHY022C201	4	0	0	4
2	Physics Lab	PHY022C211	0	0	2	1
3	Mathematics-II	MAT022C201	4	0	0	4
4	Basic Electrical Engineering	CSE022C205	3	0	0	3
5	Basic Electrical Engineering Lab	CSE022C215	0	0	2	1
6	Engineering Mechanics	CEE022C103	4	0	0	4
7	English For Technical Writing	CEN022A203	2	0	0	2
8	Design Thinking and Ideation Lab	COD022S216	0	0	2	1
	Total Credit					20

SECOND YEAR

SEMESTER III

Sl.No	Course Title	Course code	L	T	P	Credits
1.	Engineering Thermodynamics	MEE022C301	4	0	0	4
2	Strength of Materials	MEE022C303	4	0	0	4
3	Mathematics-III	MAT022C304	3	0	0	3
4	Basic Electronics Engineering	ECE022C306	4	0	0	4
5	Machine Drawing Lab	MEE022C319	1	0	4	3
6	Material Testing Lab	MEE022C315	0	0	2	1
	Total Credit					19

SEMESTER IV

Sl.No	Course Title	Course code	L	T	P	Credits
1.	Manufacturing Process I	MEE022C401	4	0	0	4
2	Engineering Materials and Application	MEE022C402	4	0	0	4
3	Fluid Mechanics	MEE022C403	4	0	0	4
4	Kinematics of Machines	MEE022C404	4	0	0	4
5	Applied Thermodynamics	MEE022C405	3	1	0	4
6	Fluid Mechanics lab	MEE022C413	0	0	2	1
	Total Credit					21

THIRD YEAR	**SEMESTER V**						
	Sl.No	**Course Title**	**Course code**	**L**	**T**	**P**	**Credits**
	1.	Machine element and system design	MEE022C501	4	0	0	4
	2	Manufacturing Process II	MEE022C502	4	0	0	4
	3	Metrology and Measurements	MEE022C503	4	0	0	4
	4	Heat and Mass Transfer	MEE022C504	4	0	0	4
	5	Heat Transfer Lab	MEE022C514	0	0	2	1
	6	Dynamics of Machines	MEE022C505	4	0	0	4
	7	Metrology and Measurements Lab	MEE022C513	0	0	2	1
	8	Dynamics of Machines Lab	MEE022C515	0	0	2	1
	Total Credit						23
	SEMESTER VI						
	Sl.No	**Course Title**	**Course code**	**L**	**T**	**P**	**Credits**
	1.	Robotics and Control	MEE022C601	4	0	0	4
	2	IC Engine	MEE022C602	4	0	0	4
	3	Computer Aided Design and Analysis	MEE022C603	4	0	1	4
	4	Tribology	MEE022C604	4	0	0	4
	5	Hydraulic Machines	MEE022D6061	4	0	0	4
	6	Gas Dynamics and Jet Propulsion	MEE022D6031	4	0	0	4
	Total Credit						24

FOURTH YEAR	**SEMESTER VII**						
	Sl.No	**Course Title**	**Course code**	**L**	**T**	**P**	**Credits**
	1.	Finite Element Analysis	MEE022C701	4	0	0	4
	2	Renewable Energy Engineering	MEE022D702	4	0	0	4
	3	Industrial Engineering	MEE022D703	4	0	0	4
	4	Internship Evaluation (Seminar)	MEE022C706	0	0	0	1
	5	Project-1	MEE022C717	0	0	0	5
	Total Credit						18
	SEMESTER VIII						
	Sl.No	**Course Title**	**Course code**	**L**	**T**	**P**	**Credits**
	1	Refrigeration & Air Conditioning	MEE022C801	4	0	0	4
	2	Computational Fluid Dynamics and Heat Transfer	MEE022D8021	4	0	0	4
	3	Tool Engineering	MEE022D8031	4	0	0	4
	4	Project-2	MEE022C804	0	0	0	5
	5	Comprehensive Viva	MEE022C837	0	0	0	2
	Total Credit						19
		TOTAL CREDITS (I, II, III, IV, V, VI, VII, VIII):					**163**

DETAILED SYLLABUS OF 1st SEMESTER

Paper I/Subject Name: Chemistry		**Subject Code:** CHY022C101
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:
The objectives of the course are to acquaint the students with the basic phenomenon/concepts of chemistry, the student faces during course of their study in the industry and Engineering field and to understand the new developments and breakthroughs efficiently in engineering and technology.

Prerequisites: Concepts of +2 level Chemistry

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Interpret periodic properties such as ionization potential, electronegativity, oxidation states, electronegativity and bulk properties and processes using thermodynamic considerations	**BT 2**
CO 2	Experiment with major chemical reactions that are used in the synthesis of molecules.	**BT 3**
CO 3	Analyze microscopic chemistry in terms of atomic and molecular orbitals and intermolecular	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	**Atomic and Molecular Structure**	Schrodinger equation. Particle in a box solution and their applications for conjugated molecules and nanoparticles. Forms of the hydrogen atom wave functions and the plots of these functions to explore their spatial variations. Molecular orbitals of diatomic molecules and plots of the multicenter orbitals. Equations for atomic and molecular orbitals. Energy level diagrams of diatomic. Pi-molecular orbitals of butadiene and benzene and aromaticity. Crystal field	**10**

II.	**Spectroscopic Techniques and Applications, Intermolecular Forces and Potential Energy Surfaces**	Principles of spectroscopy and selection rules. Electronic spectroscopy. Fluorescence and its applications in medicine. Vibrational and rotational spectroscopy of diatomic molecules. Applications. Nuclear magnetic resonance and magnetic resonance imaging, surface characterization techniques. Diffraction and scattering.	**10**

		Ionic, dipolar and van Der Waals interactions. Equations of state of real gases and critical phenomena. Potential energy surfaces of H3, H2F and HCN and trajectories on these surfaces.	
III.	**Use of free Energy in Chemical Equilibria and Periodic Properties**	Thermodynamic functions: energy, entropy and free energy. Estimations of entropy and free energies. Free energy and EMF. Cell potentials, the Nernst equation and applications. Acid base, oxidation reduction and solubility equilibria. Water chemistry. Corrosion. Use of free energy considerations in metallurgy through Ellingham diagrams. Effective nuclear charge, penetration of orbitals, variations of s, p, d and f orbital energies of atoms in the periodic table, electronic configurations, atomic and ionic sizes, ionization energies, electron affinity and electronegativity, polarizability, oxidation states, coordination numbers and geometries, hard soft acids and bases, molecular geometries	**10**

IV.	**Stereochemistry, Organic Reactions and Synthesis of a Drug Molecule**	Representations of 3 dimensional structures, structural isomers and stereoisomers, configurations and symmetry and chirality, enantiomers, diastereomers, optical activity, absolute configurations and conformational analysis. Isomerism in transitional metal compounds Introduction to reactions involving substitution, addition, elimination, oxidation, reduction, cyclization and ring openings. Synthesis of a commonly used drug molecule	**15**
		TOTAL	**45**

Chemistry Lab Syllabus

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Determination of surface tension and viscosity.
2. Thin layer chromatography.
3. Ion exchange column for removal of hardness of water.
4. Determination of chloride content of water.
5. Colligative properties using freezing point depression.
6. Determination of the rate constant of a reaction.
7. Determination of cell constant and conductance of solutions.
8. Potentiometry - determination of redox potentials and EMFs.
9. Synthesis of a polymer/drug.
10. Saponification/acid value of an oil.
11. Chemical analysis of a salt.
12. Lattice structures and packing of spheres.
13. Models of potential energy surfaces.
14. Chemical oscillations- Iodine clock reaction.
15. Determination of the partition coefficient of a substance between two immiscible liquids.
16. Adsorption of acetic acid by charcoal.
17. Use of the capillary viscosimeters to the demonstrate of the isoelectric point as the pH of minimum viscosity for gelatin sols and/or coagulation of the white part of egg.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

- A Textbook of Physical Chemistry, Negi A.S. and Anand S.C., 2nd Edition, 2007, New Age International
- Concise Inorganic Chemistry, Lee J.D., 5th Edition, 2008, John Wiley and Sons Ltd.

Reference Books:

- Atkins, P.W. and Paula, J. De,Physical Chemistry, 10th Edition, 2014, Oxford University Press
- Huheey, J.E. Keiter, E.A. Keiter, R.L Medhi, O.K.,Inorganic Chemistry Principles of Structure and Reactivity, 4th Edition, 2006, Pearson Education

Additional Readings:

- Organic Chemistry: Structure and Function by K. P. C. Volhardt and N. E. Schore, 5th Edition, http://bcs.whfreeman.com/vollhardtschore5e/default.asp
- NPTEL Course on Chemistry - I by Prof. Mangala Sunder Krishnan, IITM

Paper II/Subject Name: Mathematics-I		Subject Code: MAT022C101
L-T-P-C – 3-1-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to enable students to achieve conceptual understanding and to retain the best traditions of traditional calculus.

Prerequisites: level Concepts of class 12

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the applications of differential and integral calculus in different fields of Engineering.	**BT 2**
CO 2	Apply the single and multivariable differential and Integral calculus in engineering problems.	**BT 3**
CO 3	Analyze and assess the patterns in series	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	Basic Calculus:	Curvature, evolutes and involutes; Evaluation of definite and improper integrals; Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions. Rolle's Theorem, Mean value theorems and applications; Extreme values of functions; Linear approximation; Indeterminate forms and L'Hospital's rule.	10
II	Sequences and Series:	Limits of sequence of numbers, Calculation of limits, Infinite series; Tests for convergence; Power series, Taylor and Maclaurin series; Taylor theorem, convergence of Taylor series, error estimates.	20
III	Multivariable Calculus (Differentiation):	Limit, continuity and partial derivatives, directional derivatives, gradient, total derivative; Tangent plane and normal line; Maxima, minima and saddle points; Method of Lagrange multipliers.	10
IV	Multivariable Calculus (Integration):	Multiple Integration: Double integrals (Cartesian), change of order of integration in double integrals, Change of variables (Cartesian to polar), Applications: areas and volumes, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian), orthogonal curvilinear coordinates, Simple applications involving cubes, sphere and rectangular parallelepipeds; Scalar line integrals, vector line integrals, scalar surface integrals, vector surface integrals, Gradient, curl and divergence, Theorems of Green, Gauss and Stokes.	20
		TOTAL	60

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	1*30 NCH = 30 NCH	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. A text book of Engineering Mathematics, Bali N. P. and Narayan Iyenger N., 9th Edition, 2016, Laxmi Publication.
2. Mathematical Methods for Physics and Engineering: A Comprehensive Guide, K. F. Riley, M. P. Hobson, 3rd Edition, 2006, Cambridge University Press
3. Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
4. Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.

Reference Books:

1. Grewal B. S., Higher Engineering Mathematics, 43rd Edition, 2014, Khanna Publishers.
2. Das B. C. & Mukherjee B. N., Differential Calculus, 55th Edition, U. N. Dhur & Sons Pvt. Ltd.
3. Das B. C. & Mukherjee B. N, Integral Calculus, 57th Edition, U. N. Dhur & Sons Pvt. Ltd
4. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, John Wiley & Sons, 2006.
5. Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
6. W. E. Boyce and R. C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th Edn., Wiley India, 2009.
7. D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
8. S. L. Ross, Differential Equations, 3rd Ed., Wiley India, 1984.
9. E. A. Coddington, An Introduction to Ordinary Differential Equations, Prentice Hall India, 1995.
10. E. L. Ince, Ordinary Differential Equations, Dover Publications, 1958.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Mathematics-I.
2. https://www.vidyalankar.org/gate/assets/docs/notes/maths.pdf

Paper III/Subject Name: Fundamentals of Programming		Subject Code: CSE022C104
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: T

Objective:

The objectives of the course are to make the students capable of using C programming to solve basic as well as advanced computing problems.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:

SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Demonstrate the working of C programming language.	BT 2
CO 2	Apply the programming concepts to solve various problems.	BT 3
CO 3	Analyze and debug the errors while writing the programs.	BT 4
CO 4	Assess and design a new algorithm to solve a new real life problem.	BT 5

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Fundamentals of Programming**	Introduction to Programming; Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.) Idea of Algorithm: steps to solve logical and numerical problems. Representation of Algorithm: Flowchart/Pseudocode with examples. From algorithms to programs; source code, variables (with data types) variables and memory locations, Syntax and Logical Errors in compilation, object and executable code.	**10**
II	**Expressions, Conditional Operators and Loops**	Arithmetic expressions and precedence. Conditional Branching and Loops. Writing and evaluation of conditionals and consequent branching. Iteration and loops. Arrays, Arrays (1-D, 2-D), Character arrays and Strings	**10**

III	**Functions, Recursion, Sorting**	Basic Algorithms, Searching, Basic Sorting Algorithms (Bubble, Insertion and Selection), Finding roots of equations, notion of order of complexity through example programs (no formal definition required). Function, Functions (including using built in libraries), Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference. Recursion, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc. Quick sort or Merge sort	**10**
IV	**Advanced Programming Concepts using C**	Structures, Defining structures and Array of Structures, Pointers, Idea of pointers, Defining pointers, Use of Pointers in self-referential structures, notion of linked list (no implementation), File handling.	**15**
TOTAL			**45**

Fundamentals of Programming Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Character set, Tokens, Keywords and Identifiers, Constants, variables, data types, statements, comments, declaration of storage class, assigning values to variables.
2. Managing I/O, reading and writing characters, formatted Input/output.
3. Arithmetic operators, relational operators, logical operators, assignment operators, increment s decrement operators, conditional operators, bitwise operators, special operators.
4. Importance of decision making, decision making with if statement, if-else statement, nested if-else statements, switch-case statement.
5. Importance of iterative statements, the while statement, do-while statement, for statement, nested for looping.
6. Significance of Arrays, creation and use of one s two-dimensional arrays
7. Declaration and use of string variables, reading and writing strings.
8. Benefits of user-defined functions, creation and use of user-defined functions, parameter passing, return types.
9. Use of Pointers, declaration s initialization of pointer variables, accessing a variable through its pointer.
10. Defining, opening s closing files in C.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	1*30 NCH = 30 NCH	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Computer Fundamentals and Programming in C,* Reema Thareja, 2nd Edition, 2016, Oxford University Press, Delhi.

Reference Books:

1. E Balaguruswamy, *Computing Fundamentals and C Programming*, 1st Edition, 2017, McGraw Hill.
2. Venugopal and Prasad, *Mastering C*, 2nd Edition, 2017, Tata McGraw Hill.
3. Yashawant Kanetkar,*Let us C,* 15th Edition, 2017, BPB.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Programming%20for%20Problem%20Solving.pdf
2. NPTEL course on Introduction to Programming in C by Prof. Satyadev Nandakumar, IIT, Kanpur
3. NPTEL course on Problem Solving Through Programming in C by Prof. Anupam Basu, IIT, Kharagpur

Subject Name: Engineering Graphics & Design Lab		Subject Code: CEE022C114
L-T-P-C – 0-0-4-2	Credit Units: 02	Scheme of Evaluation: P

Objective:

The objectives of the course are to make students understand the process of drawing projections and sections and basic engineering drawing formats and to convert sketches to engineered drawings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Relate** with the concepts of drawings and projections	**BT 1**
CO 2	**Understand** the dimension and figures using the drawing instruments and acquire visualisation skills, projection of	**BT 2**
CO 3	**Utilize** engineering curves in tracing the paths of simple machine components.	**BT 3**
CO 4	**Analyse** and assess sketches to convert them to engineered drawings.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 60 (4 hours per week)
Minimum 10 Laboratory experiments based on the following-
Objective: To determine and verify different parameter of Fluid

Experiment	No.
To prepare a sheet on Lettering, dimensions & Scales	01
To prepare different types scales used in Engineering Drawings	02
To prepare curves used in Engineering Graphics	03
To prepare a sheet on Projection of point & lines	04
To prepare a sheet on Projection of planes	05
To prepare a sheet on Projection of solids	06
To prepare a sheet on orthographic & Isometric projections	07
To draw basic geometrical entities using Autocad software	08
To draw Orthographic projections in Autocad Software	09
To draw Isometric projections in Autocad Software	10

Text Books:

1. Engineering Drawing; Bhatt, N.D, 53rd Edition, 2016, Charotar Publishing House

Reference Books:

1. Jolhe Dhananjay A; Engineering drawing, 5th Edition, 2010, Tata McGraw-Hill Education Pvt. Ltd., New Delhi

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Engineering%20Graphics%20Manual%20final.pdf
2. https://www.pvpsiddhartha.ac.in/autonomus14/1-1/it/IT1L3.pdf
3. NPTEL Course on Engineering Drawing and Computer Graphics by Prof. Rajaram Lakkaraju, IIT, Kharagpur
4. NPTEL Course on Engineering Graphics by Prof. Nihar Ranjan Patra, IIT, Kanpur

Paper V/Subject Name: Manufacturing Practices Workshop		Subject Code: MEE022C115
L-T-P-C – 0-0-4-2	**Credit Units: 02**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide exposure to the students with hands-on experience on various basic engineering practices in Civil, Mechanical, Electrical and Electronics Engineering.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the different manufacturing processes which are commonly employed in the industry	**BT 2**
CO 2	**Utilize** tools, instruments and techniques learnt to perform basic household chores in terms of house wiring, carpentry etc	**BT 3**
CO 3	**Experiment** using the tools and techniques learnt for various purposes and decide on the best prospect.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 40 (4 hours per week)

Minimum 10 Laboratory experiments based on the following-

The lecture sessions will be on the following topics:

- Manufacturing Methods- casting, forming, machining, joining, advanced manufacturing methods.
- CNC machining, Additive manufacturing.
- Fitting operations s power tools.
- Electrical s Electronics.
- Carpentry.
- Plastic moulding, glass cutting.
- Metal casting.
- Welding (arc welding s gas welding), brazing g topics:

And the lab sessions will on the topics:

- Machine shop
- Fitting shop
- Carpentry
- Electrical s Electronics
- Welding shop (Arc welding + Gas welding)
- Casting
- Smithy
- Plastic moulding s Glass Cutting

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
-	2*20 NCH = 40 NCH	20 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. Elements of Workshop Technology, Hajra Choudhury, S K, Hajra Choudhury, A K, 14th Edition, 2007, Mumbai Media Promoters
2. Manufacturing Technology – I, Gowri P. Hariharan and A. Suresh Babu, 2008, Pearson Education.

Reference Books:

1. Roy A. Lindberg, Processes and Materials of Manufacture", 4th Edition, 1668, Prentice Hall India,

Additional Readings:

1. http://mm- coep.vlabs.ac.in/LaserSpotWelding/Theory.html?domain=Mechanical%20Engineeringslab=Welcome%20to%20Micromachining%20laboratory
2. http://fab-coep.vlabs.ac.in/exp7/Theory.html?domain=Mechanical%20Engineeringslab=Welcome%20to%20FAB%20laboratory

Paper VI/Subject Name: Universal Human Values		Subject Code: BHS022A103
L-T-P-C – 2-0-0-2	Credit Units: 02	Scheme of Evaluation: T

Objective:

The objectives of the course are to help the students appreciate the essential complementarily between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the importance of following the basic universal human values.	**BT 2**
CO 2	**Apply** the holistic understanding in one's day-today life so as to keep oneself happy and to socialize with nature, society, etc	**BT 3**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Value Education**	Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education), Understanding Value Education, Sharing about Oneself, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Exploring Human Consciousness, Happiness and Prosperity – Current Scenario, Lectured, Method to fulfil the Basic Human Aspirations, Exploring Natural Acceptance	**11**
II	**Harmony in Human Being**	Understanding Human being as the Co-existence of the Self and the Body, distinguishing between the Needs of the Self and the Body, Exploring the difference of Needs of Self and Body, The Body as an Instrument of the Self Understanding Harmony in the Self, Exploring Sources of Imagination in the Self, Harmony of the Self with the Body, Programme to ensure self-regulation and Health, Exploring Harmony of Self with the Body	**11**
III	**Harmony in the Family s Society**	Harmony in the Family – the Basic Unit of Human Interaction, 'Trust' – the Foundational Value in Relationship", Exploring the Feeling of Trust, 'Respect' – as the Right Evaluation, Exploring the Feeling of Respect, Other Feelings, Justice in Human-to- Human Relationship Understanding Harmony in the Society, Vision for the Universal Human Order, Exploring Systems to fulfil Human Goal	**11**

IV	**Harmony in Nature s Implications of Holistic Understanding**	Understanding Harmony in the Nature, Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature, Exploring the Four Orders of Nature, Realizing Existence as Co-existence at All Levels, The Holistic Perception of Harmony in Existence, Exploring Co-existence in Existence Natural Acceptance of Human Values, Definitiveness of (Ethical) Human Conduct, Exploring Ethical Human Conduct, A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order, Competence in Professional Ethics, Exploring Humanistic Models in Education, Holistic Technologies, Production Systems and Management Models- Typical Case Studies, Strategies for Transition towards Value- based Life and Profession Exploring Steps of Transition towards Universal Human Order	**11**
		TOTAL	**44**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2 * 22 NCH = 44 NCH	-	8 * 2 NCH = 16 NCH (Seminar, Case Study, Discussion, Internship)

Text Books:

1. *A Foundation Course in Human Values and Professional Ethics,* R R Gaur, R Asthana, G P Bagaria, 2nd Revised Edition, Excel Books, New Delhi

Reference Books:

1. *Human Values*, A.N. Tripathi, 3rd Edition, 2016, New Age Intl. Publishers, New Delhi,

Additional Readings:

1. https://uhv.org.in/uhv2notes
2. https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php

Paper VII/Subject Name: Sports and Yoga		Subject Code: CEE022S117
L-T-P-C – 0-0-2-1	Credit Units: 01	Scheme of Evaluation: P

Objective:

The objectives of the course are to make the students understand the importance of sound health and fitness principles as they relate to better health, to expose the students to a variety of physical and yogic activities aimed at stimulating their continued inquiry about Yoga, physical education, health and fitness and to develop among students an appreciation of physical activity as a lifetime pursuit and a means to better health.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** basic skills associated with yoga and physical activities including strength and flexibility, balance and coordination	**BT 2**
CO 2	**Experiment with** different forms of yoga to keep oneself physically fit and mentally strong	**BT 3**
CO 3	**Assess** current personal fitness levels	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Physical Education, Olympic Movement, Fitness, Wellness s Lifestyle**	Meaning s definition of Physical Education. Aims s Objectives of Physical Education. Changing trends in Physical Education, Ancient s Modern Olympics (Summer s Winter), Olympic Symbols, Ideals, Objectives s Values, Awards and Honours in the field of Sports in India (Dronacharya Award, Arjuna Award, Dhayanchand Award, Rajiv Gandhi Khel Ratna Award etc., Meaning s Importance of Physical Fitness s Wellness, Components of Physical fitness, o Components of Health-related fitness, Components of wellness, Preventing Health Threats through Lifestyle Change, Concept of Positive Lifestyle.	**5**

II	**Anatomy s Physiology in Physical Education, Sports,** **Yoga s Postures**	Define Anatomy, Physiology s Its Importance, ffect of exercise on the functioning of Various Body Systems. (Circulatory System, Respiratory System, Neuro-Muscular System etc.), Meaning s Importance of Kinesiology s Biomechanics in Physical Edu. s Sports, o Newton's Law of Motion s its application in sports. o Friction and its effects in Sports, Meaning and Concept of Postures, Causes of Bad Posture. Advantages s disadvantages of weight training. Concept s advantages of Correct Posture. Common Postural Deformities – Knock Knee; Flat Foot; Round Shoulders; Lordosis, Kyphosis, Bow Legs and Scoliosis. Corrective Measures for Postural Deformities	**5**

III	**Yoga s Lifestyle**	Meaning s Importance of Yoga, Elements of Yoga, Asanas, Pranayama, Meditation s Yogic Kriyas, yoga for concentration s related Asanas (Sukhasana; Tadasana; Padmasana s Shashankasana) Relaxation Techniques for improving concentration - Yog-nidra, Asanas as preventive measures. Hypertension: Tadasana, Vajrasana, Pavan Muktasana, Ardha Chakrasana, Bhujangasana, Sharasana. Obesity: Procedure, Benefits s contraindications for Vajrasana, Hastasana, Trikonasana, Ardh Matsyendrasana. Back Pain: Tadasana, Ardh Matsyendrasana, Vakrasana, Shalabhasana, Bhujangasana. Diabetes: Procedure, Benefits s contraindications for Bhujangasana, Paschimottasana, Pavan Muktasana, Ardh Matsyendrasana. Asthema: Procedure, Benefits s contraindications for Sukhasana, Chakrasana, Gomukhasana, Parvatasana, Bhujangasana, Paschimottasana, Matsyasana.	**5**
IV	**Training, Planning and Psychology in Sports**	Meaning of Training, Warming up and limbering down, Skill, Technique s Style, Meaning and Objectives of Planning. Tournament – Knock-Out, League/Round Robin s Combination. Definition s Importance of Psychology in Physical Edu. s Sports, Define s Differentiate Between Growth s Development, Adolescent Problems s Their Management, Emotion: Concept, Type s Controlling of emotions, Meaning, Concept s Types of Aggressions in Sports. Psychological benefits of exercise. Anxiety s Fear and its effects on Sports Performance. Motivation, its type s techniques. Understanding Stress s Coping Strategies	**5**
		TOTAL	**20**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
	1*20 NCH = 20 NCH	10 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *Modern Trends and Physical Education*, Ajmer Singh, Gill J.S, Bains J, 4th Edition, 2012, Kalyani Publishers

Reference Books:

1. B.K.S. Iyengar, *Light on Yoga*, 2006, Thorsons

DETAILED SYLLABUS OF 2ND SEMESTER

Paper I/Subject Name: Physics		Subject Code: PHY022C201
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: TP

Objective:

The objectives of the course are to make the students enhance the fundamental knowledge in Physics and its applications relevant to various streams of Engineering and Technology

Prerequisites: Concepts of Physics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the basic concepts of Physics	**BT 2**
CO 2	**Identify** the applications of Physics in technical field.	**BT 3**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Classical Mechanics and Dynamics**	Transformation of scalars and vectors under Rotation transformation; Forces in Nature; Newton's laws and its completeness in describing particle motion; Form invariance of Newton's Second Law; Solving Newton's equations of motion in polar coordinates; Problems including constraints and friction; Extension to cylindrical and spherical coordinates. Potential energy function; F = - Grad V, equipotential surfaces and meaning of gradient; Conservative and non- conservative forces, curl of a force field; Central forces; Conservation of Angular Momentum; Energy equation and energy diagrams; Elliptical, parabolic and hyperbolic orbits; Kepler problem; Application: Satellite manoeuvres.	**10**
II.	**Advanced Dynamics and Oscillatory Motion**	Non-inertial frames of reference; Rotating coordinate system: Five-term acceleration formula. Centripetal and Coriolis accelerations; Applications: Weather systems, Foucault pendulum; Harmonic oscillator; Damped harmonic motion – over-damped, critically damped and lightly- damped oscillators; Forced oscillations and resonance.	**10**
III.	**Rigid Body Dynamics and Kinematics**	Definition and motion of a rigid body in the plane; Rotation in the plane; Kinematics in a coordinate system rotating and translating in the plane; Angular momentum about a point of a rigid body in planar motion; Euler's laws of motion, their independence from Newton's laws, and their necessity in describing rigid body motion; Examples.	**15**

IV	**Advanced Rigid Body Dynamics: Three-Dimensional Motion**	Introduction to three-dimensional rigid body motion — only need to highlight the distinction from two- dimensional motion in terms of (a) Angular velocity vector, and its rate of change and (b) Moment of inertia tensor; Three-dimensional motion of a rigid body wherein all points move in a coplanar manner: e.g. Rod executing conical motion with center of mass fixed — only need to show that this motion looks two-dimensional but is three-dimensional, and two-dimensional formulation fails.	**10**
		TOTAL	**45**

Physics Lab Syllabus

Detailed Syllabus:

Experiment	Experiment Title	Lab Hours
I	Determination of Moment of Inertia of a given solid about its own axis by using M.I.Table	2
II	Determination of Young's Modulus using Searle's Apparatus	2
III	Determination of Rigidity of Modulus of the material of the given rod by Statiscal method	2
IV	Determination of Powers of Given lenses using an Optical Bench i. Concave Lens, ii Convex Lens	2
V	Determination of Resistance of a Galvanometer using Post Office Box.	2
VI	To determine the mechanical equivalent of heat by Joule's calorimeter	2
VII	Determination of ratio of E.M.F of two cells using Potentiometer.	2
VIII	To determination of the focal length of a convex mirror with the help of an auxiliary lens.	2
IX	Determination of Horizontal Components of Earth's Magnetic field using Magnetometer	2
X	Determination of coefficient of Viscosity of water by Capillary Flow Method	2

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. Engineering Mechanics, 2nd ed. – D.S. Bedi, M.P. Poonia
2. Basic Mechanical Engineering – S.C. Sharma, M.P. Poonia
3. Engineering Mechanics, 2nd ed. — MK Harbola
4. Introduction to Mechanics — MK Verma
5. An Introduction to Mechanics — D Kleppner & R Kolenkow
6. Principles of Mechanics — JL Synge & BA Griffiths
7. Mechanics — JP Den Hartog
8. Engineering Mechanics - Dynamics, 7th ed. - JL Meriam
9. Mechanical Vibrations — JP Den Hartog
10. Theory of Vibrations with Applications — WT Thomson

Reference Books:

1. Singh A.K. and Malik Hitendra Engineering Physics, 2nd Edition, 2016, McGraw Hill Education Private Limited. New Delhi.
2. Gaur R.K and Gupta S.L, Engineering Physics, 2015, Dhanpat Rai publication, New Delhi.
3. Arthur Beiser, Shobhit Mahajan, S. Rai. Choudhury, Concept of Modern physics, 6th Edition, 2006, McGraw-Hill education Private limited. New Delhi.
4. M Ghosh s D Bhattacharya, A Textbook of Oscillations, Waves and Acoustics, 5th Edition, 2016, S. Chand publication.

Additional Readings

1. https://www.griet.ac.in/nodes/Engineering%20Physics%20Notes.pdf
2. https://mrcet.com/downloads/digital_notes/HS/R20/Engineering%20Physics.pdf
3. NPTEL Course on Introduction To Electromagnetic Theory by Prof. Manoj Harbola, IIT Kanpur
4. NPTEL Course on Engineering Mechanics by Prof. Manoj Harbola, IIT Kanpur

Alternative NPTEL/SWAYAM Course:

S. No.	NPTEL Course Name	Instructor	Host Institute
1	ENGINEERING MECHANICS	PROF. MANOJ HARBOLA	IIT KANPUR

Paper II/Subject Name: Mathematics II		Subject Code: MAT022C201
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to teach the students Mathematics fundamentals necessary to formulate, solve and analyze engineering problems

Prerequisites: Concepts of Mathematics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** essential tool of calculus, multivariate differentiation and integration.	**BT 2**
CO 2	**Utilize** the essential tools in the field of applied sciences and related fields.	**BT 3**
CO 3	**Analyze** and **evaluate** the qualitative behavior of solutions of systems of differential equations and interpret in the context of an underlying model.	**BT 4 s 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Basic Calculus:**	Curvature, evolutes and involutes; Evaluation of definite and improper integrals; Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions. Rolle's Theorem, Mean value theorems and applications; Extreme values of functions; Linear approximation; Indeterminate forms and L'Hospital's rule.	**10**
II	**Sequences and Series:**	Limits of sequence of numbers, Calculation of limits, Infinite series; Tests for convergence; Power series, Taylor and Maclaurin series; Taylor theorem, convergence of Taylor series, error estimates.	**20**
III	**Multivariable Calculus (Differentiation):**	Limit, continuity and partial derivatives, directional derivatives, gradient, total derivative; Tangent plane and normal line; Maxima, minima and saddle points; Method of Lagrange multipliers.	**10**

IV	**Multivariable Calculus (Integration):**	Multiple Integration: Double integrals (Cartesian), change of order of integration in double integrals, Change of variables (Cartesian to polar), Applications: areas and volumes, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian), orthogonal curvilinear coordinates, Simple applications involving cubes, sphere and rectangular parallelepipeds; Scalar line integrals, vector line integrals, scalar surface integrals, vector surface integrals, Gradient, curl and divergence, Theorems of Green, Gauss and Stokes.	**20**
TOTAL			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 15 NCH = 60 NCH	2 * 15 NCH = 30 NCH	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. Reena Garg, Engineering Mathematics, Khanna Book Publishing Company, 2022.
2. Reena Garg, Advanced Engineering Mathematics, Khanna Book Publishing Company, 2021.
3. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 6th Edition, Pearson, Reprint, 2002.
4. Erwin Kreyszig, Advanced Engineering Mathematics, 6th Edition, John Wiley s Sons, 2006.
5. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
6. Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
7. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
8. A text book of Engineering Mathematics, Bali N. P. and Narayan Iyenger N., 6th Edition, 2016, Laxmi Publication.
9. Mathematical Methods for Physics and Engineering: A Comprehensive Guide, K. F. Riley, M. P. Hobson, 3rd Edition, 2006, Cambridge University Press

Reference Books:

1. Grewal B. S., *Higher Engineering Mathematics,* 43rd Edition, 2014, Khanna Publishers.
2. Raisinghania M.D., *Ordinary and Partial Differential Equations*, 17th Edition, 2014, S. Chand and Co., New Delhi.
3. Narayna S., *A Text Book of Vector Calculus,* Revised Edition, 2006, S. Chand s Co., New Delhi.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/R-18%20Mathematics-II.pdf
2. http://www.bosecuttack.in/studentcorner/LECTURE_NOTE.MATH2.2ND_SEM__1_.pdf
3. https://www.srividyaengg.ac.in/coursematerial/Iyear/111223.pdf

Paper III/Subject Basic Electrical Engineering		Subject Code: CSE022C205
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: TP

Objective:

The objectives of the course are to make students understand the basic electrical terminologies and familiarize them with the basic concepts of D.C., single-phase and three-phase A.C. networks.

Prerequisites: Basic concepts of D.C. networks of Class XII, Electromagnetic Induction and A.C. Fundamentals etc.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the concept behind basic electric and magnetic circuits.	**BT 2**
CO 2	**Apply** the working principles of electrical machines and power converters in real-life.	**BT 3**

Detailed Syllabus:

Modules	Topic	Course Content	Hours
I.	**DC Circuits**	D. C. Circuits covering, Ohm's Law and Kirchhoff's Laws; Analysis of series, parallel and series-parallel circuits excited by independent voltage sources; Power and energy; Electromagnetism covering, Faradays Laws, Lenz's Law, Fleming's Rules, Statically and dynamically induced EMF; Concepts of self-inductance, mutual inductance and coefficient of coupling; Energy stored in magnetic fields	**15**
II.	**AC Circuits**	Single Phase A.C. Circuits covering, Generation of sinusoidal voltage-definition of average value, root mean square value, form factor and peak factor of sinusoidal voltage and current and phasor representation of alternating quantities; Analysis with phasor diagrams of R, L, C, RL, RC and RLC circuits; Real power, reactive power, apparent power and power factor, series, parallel and series- parallel circuits; Three Phase A.C. Circuits covering, Necessity and Advantages of three phase systems, Generation of three phase power, definition of Phase sequence, balanced supply and balanced load; Relationship between line and phase values of balanced star and delta connections; Power in balanced three phase circuits, measurement of power by two wattmeter method;	**15**

III.	**Electrical Machines:**	Principle of operation and construction of single-phase transformers (core and shell types). EMF equation, losses, efficiency and voltage regulation; Synchronous Generators covering, Principle of operation; Types and constructional features; EMF equation; working principle of DC machine as a generator and a motor; Types and constructional features; EMF equation of generator, relation between EMF induced and terminal voltage enumerating the brush drop and drop due to armature reaction; DC motor working principle; Back EMF and its significance, torque equation; Types of D.C. motors, characteristics and applications; Necessity of a starter for DC motor	**10**
IV.	**Electrical Installations:**	Three Phase Induction Motors covering; Concept of rotating magnetic field; Principle of operation, types and constructional features; Slip and its significance; Applications of squirrel cage and slip ring motors; Necessity of a starter, star-delta starter. Sources of Electrical Power covering, Introduction to Wind, Solar, Fuel cell, Tidal, Geothermal, Hydroelectric, Thermal-steam, diesel, gas, nuclear power plants; Concept of cogeneration, and distributed generation;	**5**
		TOTAL	**45**

Basic Electrical Engineering Lab Syllabus

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

Lab	Experiments	Hours
I	To verify Thevenin's Theorem for DC network	**2**
II	To verify Maximum Power Transfer Theorem for DC network	**2**
III	Study of R-L-C Series circuit and determine R,L,C, cos Φ,P and Q and draw the phasor diagram	**2**
IV	Study of R-L-C Parallel circuit and determine R,L,C,cos Φ,P and Q and draw the phasor diagram	**2**
V	Calibration of a milli-ammeter as a voltmeter.	**2**
VI	To determine the ohmic and effective resistance (armature winding)	**2**
VII	To study the characteristics of a filament lamp	**2**
VIII	To measure the power in a single-phase load using one wattmeter	**2**
IX	To measure the insulation resistance using Megger	**2**
X	Demonstration of house wiring	**2**
	TOTAL	**20**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. A Text Book of Electrical Technology, Thereja, B.L., 1st Edition revised, 2008, S Chand s Company Ltd. Ram Nagar; New Delhi.
2. Basic Electrical Engineering, D. P. Kothari, I. J.Nagrath, 3rd Edition, 2006, Tata McGraw-Hill

Reference Books:

1. D. C. Kulshreshtha, *Basic Electrical Engineering*, 1st Edition, 2006, McGraw-Hill
2. E. Hughes, *Electrical and Electronics Technology,* 10th Edition, 2011, Pearson Publication

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Basic%20Electrical%20Engineering%20R-20.pdf
2. https://www.cet.edu.in/noticefiles/231_BASIC_ELECTRICAL_ENGG-min.pdf
3. NPTEL Course on Basic Electrical Circuits by Prof. Nagendra Krishnapura, IITM
4. NPTEL Course on Fundamentals of Electrical Engineering by Prof. Debapriya Das, IIT, Kharagpur

Paper IV/Subject Name: English for Technical Writing		Subject Code: CEN022A201
L-T-P-C – 2-0-0-2	Credit Units: 02	Scheme of Evaluation: T

Objective:

The objectives of the course are to provide learning environment to practice listening, speaking, reading and writing skills, to assist the students to carry on the tasks and activities through guided instructions and materials and to effectively integrate English language learning with employability skills and training.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** basic proficiency in English.	**BT 2**
CO 2	**Develop** reading and listening comprehension, writing and speaking skills.	**BT 3**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	Vocabulary Building	The concept of Word Formation, Root words from foreign languages and their use in English, Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives, Synonyms, antonyms, and standard abbreviations.	10
II	Basic Writing Skills	Sentence Structures, Use of phrases and clauses in sentences, Importance of proper punctuation, creating coherence, organizing principles of paragraphs in documents, Techniques for writing precisely, Subject-verb agreement, Noun-pronoun agreement, Misplaced modifiers, Articles, Prepositions, Redundancies, Clichés	10
III	Writing Practices	Nature and Style of sensible Writing, Describing, Defining 1.3. Classifying, providing examples or evidence, Writing introduction and conclusion, Comprehension, Précis Writing, Essay Writing.	10
IV	Oral Communication	Listening Comprehension, Pronunciation, Intonation, Stress and Rhythm, Common Everyday Situations: Conversations and Dialogues, Communication at Workplace, Interviews, Formal Presentations	10
TOTAL			40

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2*20 NCH = 40 NCH	-	20 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

- Effective Communication Skills. Kul Bhushan Kumar, 2022, Khanna Book Publishing
- Practical English Usage, Michael Swan. 1665, OUP

Reference Books:

- F.T. Wood, Remedial English Grammar., 2007, Macmillan.
- William Zinsser, On Writing Well, 2001, Harper Resource Book.
- Liz Hamp-Lyons and Ben Heasly, Study Writing, 2006,
- Sanjay Kumar and PushpLata, Communication Skills, 2011, Oxford University Press.

Additional Readings:

- AICTE's Prescribed Textbook: English (with Lab Manual), Khanna Book Publishing Co., https://khannabooks.com/index.php?route=product/productspath=66_105sproduct_id=480
- NPTEL Course on English Language for Competitive Exams by Prof. by Aysha Iqbal, IIT, Madras
- NPTEL Course on Technical English for Engineers by Prof. by Aysha Iqbal, IIT, Madras

Paper-V Subject : Design Thinking and Ideation Lab	**Subject Code:** COD022S216
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

To develop students' creative thinking, innovation, and problem-solving abilities through the Design Thinking process, enabling them to identify real-world engineering challenges, generate sustainable solutions, develop prototypes, and communicate innovative ideas effectively.

Prerequisites: None

Course Outcomes:

Upon successful completion of the course, students will be able to:

CO	Course Outcome	Bloom's Level
CO1	Explain the principles of design thinking, creativity, ideation and innovation in solving engineering problems.	BT1–BT2
CO2	Apply empathy, stakeholder analysis and creative thinking techniques to identify problems and generate innovative engineering concepts.	BT3
CO3	Develop, test and refine engineering prototypes by integrating sustainability, technical feasibility and user feedback.	BT4–BT5
CO4	Design, document and present innovative engineering solutions considering entrepreneurship, intellectual property and scalability.	BT5–BT6

Detailed Syllabus:

Module	Topics	Course Contents	Hours
I	**Foundations of Design Thinking, Creativity and Problem Identification**	Introduction to creativity, innovation and ideation; importance of Design Thinking in engineering; stages of Design Thinking (Empathize, Define, Ideate, Prototype, Test); learning styles and creative thinking; empathy, stakeholder analysis, field observation, need identification; understanding societal, environmental and sustainability challenges; formulation of engineering problem statements.	**5**
II	**Idea Generation and Concept Development**	Brainstorming techniques, mind mapping, SCAMPER and other creative tools; problem-solving approaches; idea screening and feasibility analysis; technical, economic and environmental considerations; customer-centric design; user experience; engineering concept development; sustainable and resilient design principles; selection of suitable materials and methods.	**5**
III	**Prototype Development, Testing and Validation**	Engineering product design process; prototype development (low-fidelity and working models); rapid prototyping techniques; testing and validation; user feedback; ergonomic considerations; iterative design improvement; group activities, teamwork and appreciation of individual differences; documentation of prototype development.	**6**

IV	**Innovation, Entrepreneurship and Final Project**	Technical documentation and report writing; presentation and pitching skills; introduction to entrepreneurship and startup ecosystem; intellectual property rights (IPR); techno-economic feasibility; scalability of engineering solutions; final prototype demonstration; peer review; solving real-life civil engineering problems using Design Thinking and innovative solutions.	**6**

Suggested Text Books

- E. Balaguruswamy, *Developing Thinking Skills (The Way to Success)*, Khanna Publishing House, 2022.
- Iñigo Cuiñas & Manuel José Fernández Iglesias, *Design Thinking for Engineering: A Practical Guide*, Institution of Engineering and Technology, 2023.
- Christian Müller-Roterberg, *Design Thinking For Dummies*, Wiley, 2020.
- Idris Mootee, *Design Thinking for Strategic Innovation*, Adams Media, 2014.
- Chris Hackett, *The Big Book of Maker Skills*, Weldon Owen, 2018.

Paper V/Subject Name: Engineering Mechanics	**Subject Code: CEE022C217**
L-T-P-C –4-0-0-4 **Credit Units: 04**	**Scheme of Evaluation: T**

Course Objective:
To introduces some basic principles of engineering mechanics in as simple a manner as possible.

Course Outcomes:

On successful completion of the course the students will be able to:

SI No	Course Outcome	Bloom's Taxonomy Level
CO 1	Understand the fundamental principles of engineering mechanics by explaining the concepts of force systems, equilibrium, free body diagrams, moments, and the laws governing the behavior of rigid bodies.	BT 2
CO 2	Apply the principles of statics to analyze equilibrium of particles and rigid bodies, determine support reactions, and solve engineering problems involving trusses, frames, centroids, and moments of inertia.	BT 3
CO 3	Analyze the effects of friction, virtual work, and energy methods to solve engineering mechanics problems involving machines, connected bodies, and structural systems under different loading conditions.	BT 4
CO 4	Evaluate the stability, safety, and mechanical behavior of engineering structures and mechanical systems by selecting appropriate analytical methods and interpreting the results for practical engineering applications.	BT 5

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Equilibrium of System of Forces**	Force Systems Basic concepts, equilibrium in 2-D & 3-D; Rigid Body equilibrium; System of Forces, Coplanar Concurrent Forces, Components in Space – Resultant-Moment of Forces and its Application; Couples and Resultant of Force System. Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems and Spatial Systems; Static Indeterminacy	**15**
II.	**Analysis Of Structure**	Two force members, Analysis of plane trusses by Method of joints Analysis of plane trusses by method of section, Analysis of plane frames, Cables subjected to point load multi force member.	**15**

III.	**Centroid, Centre Of Gravity And Moment Of Inertia**	Centroid of simple figures from first principle, Centroid of composite sections; Centre of Gravity and its implications. Area moment of inertia- Definition, Moment of inertia of plane sections from first principles, Theorems of moment of inertia, Moment of inertia of standard sections and composite sections; Mass moment inertia of circular plate, Cylinder, Cone, Sphere, Hook.	**15**
IV	**Friction, Virtual Work And Energy Method**	Types of friction, Limiting friction, Laws of Friction, Static and Dynamic Friction; Motion of Bodies, wedge friction, screw jack. Virtual displacements, principle of virtual work for particle and ideal system of rigid bodies, degrees of freedom. Active force diagram, systems with friction, mechanical efficiency. Conservative forces and potential energy (elastic and gravitational), energy equation for equilibrium. Applications of energy method for equilibrium. Stability of equilibrium.	**15**
TOTAL			**60**

Textbooks:

1. F. P. Beer and E. R. Johnston (2011), Vector Mechanics for Engineers, Vol I - Statics, Vol II, – Dynamics, 9th Ed, Tata McGraw Hill
2. R. C. Hibbler (2006), Engineering Mechanics: Principles of Statics and Dynamics, Pearson Press.
3. Timoshenko S, Young H.D. and Rao J.V, Engineering Mechanics, Tata McGraw Hill Publications.

DETAILED SYLLABUS OF 3rd SEMESTER

Paper I/Subject Name: Engineering Thermodynamics		**Subject Code: MEE022C301**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective: The objective of this course is to equip students with a fundamental understanding of thermodynamic principles, including work and heat interactions, energy balance in systems, and the application of the First Law to various energy conversion devices..

Prerequisites: Concepts of +2 level Physics & Chemistry

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**understand** the fundamental concepts of thermodynamics by explaining work and heat interactions, system properties, and the laws governing energy transfer	**BT 2**
CO 2	**apply** the First and Second Laws of Thermodynamics to analyze energy conversion devices such as engines, turbines, and refrigerators,	**BT 3**
CO 3	**evaluate** the thermodynamic properties of substances using property tables, equations of state, and phase diagrams to determine changes in	**BT4**
CO 4	**differentiate** between high-grade and low-grade energy and assess the limitations of energy conversion based on entropy and irreversibility in real-world applications	**BT 4**

Detailed Syllabus:

Modules	Topics / Course content	Hours
I.	**Introduction** : Role of Thermodynamics in Engineering and Science , Applications of Thermodynamics : Power Generation, Thermal Environment Control, Cooling of Electrical Systems and Electronic Devices, Analysis of Manufacturing Processes. **Basic Definitions and Units**- System & Control volume; Property, State & Process; Exact & Inexact differentials; Work - Thermodynamic definition of work; examples; Displacement work; Path dependenceof displacement work and illustrations for simple processes; electrical, magnetic, gravitational, spring and shaft work. **Temperature**: Definition of thermal equilibrium and Zeroth law; Temperature scales; Various Thermometers- Definition of heat; examples of heat/work interaction in systems- First Law for Cyclic & Non-cyclic processes; Concept of total energy E ; Demonstration that E is a property; Various modes of energy, Internal energy and Enthalpy.	**15**
II.	**Pure substance**: Definition of Pure Substance -- Facts about Pure Substances -- Vapor -- liquid -- solid Phase Equilibrium -- Equation of State for the Vapor Phase : Simple substance, Ideal Gas Characterization, Ideal Gas Equation, Compressibility Effects and Resulting Equations of State -- Real Gases. Definitions of saturated states; P-v-T surface; Use of steam tables: Saturation tables; Superheated tables; Identification of states & determination of properties, Mollier's chart.	**15**
III.	**Heat and Work**: Definition of Thermodynamic Work -- Units for Work -- Forms of Work – Definition of Heat -- Inter Convertibility of Heat/work into Work/heat -- Governing Principles -- Sign Convention. **First Law** for Flow Processes - Derivation of general energy equation for a control volume; Steady state steady flow processes including throttling; Examples of steady flow devices **Second law** - Definitions of direct and reverse heat engines; Definitions of thermal efficiency and COP; Kelvin-Planck and Clausius statements; Definition of reversible process; Internal and external irreversibility; Carnot cycle; Absolute temperature scale.	**15**
IV.	**Entropy :** Clausius inequality; Definition of entropy S; Demonstration that entropy S is a property; Evaluation of S for solids, liquids, ideal gases and ideal gas mixtures undergoing various processes; Principle of increase of entropy; Illustration of processes in T-s coordinates; Definition of Isentropic efficiency for compressors, turbines and nozzles. **Exergy Analysis** Definition of Exergy -- Exergy Analysis of System and Control volume -- Exergetic efficiency **Analysis of Power Generation Cycles** :Air-standard Power Cycles -- Concept -- Carnot Cycle -- Otto Cycles -- Diesel Cycle -- Dual Cycle -- Brayton Cycle -- Efficiency and Mean Effective Pressure and Temperature Vapor Power Cycles -- Concept -- Carnot Cycle -- The Rankine Cycle -- Effect of Temperature and Pressure on The Rankine Cycle -- The Superheat Cycle --The Reheat Cycle -- The Regenerative Cycle -- Deviation of Actual Cycle from Ideal Cycle.	**15**
Total		**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 15 NCH = 60 NCH		8 * 4 NCH = 32 NCH (Problem Solving, Assignments, Quiz, Presentation, Case Study, Discussion)

Text Books:

1. Cengel and Boles, Fundamentals of Thermodynamics, 4th edition, 2001, Tata McGraw HillPublication
2. P K Nag, Engineering Thermodynamics, 6th edition, 2017, McGraw Hill Education
3. Sonntag, R. E, Borgnakke, C. and Van Wylen, G. J., Fundamentals of Thermodynamics, 6thEdition, 2003, John Wiley and Sons.
4. Moran, M. J. and Shapiro, H. N., 1999, Fundamental of Engineering Thermodynamic, John Wileyand Sons.

Alternative NPTEL/SWAYAM Course

S.No.	NPTEL Course Name	Instructor	Host Institute
1	ENGINEERING THERMODYNAMICS	PROF. V. BABU	IIT MADRAS

Paper II/Subject Name: Strength of Materials **Subject Code: MEE022C303**

L-T-P-C – 4-0-0-4 **Credit Units: 04** **Scheme of Evaluation: TP**

Objective:

The course aims to teach stress-strain analysis, material behavior under loading, structural problem-solving, energy methods, stability evaluation, failure theories, and design applications for mechanical engineering.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	**Blooms Taxonomy Level**
CO 1	Understand and explain the concepts of stress and strain in deformable solids.	**Understand (BT2)**
CO 2	Analyze engineering problems involving bending and torsion to determine stresses, strains, and deformations.	**Analyze (BT4)**
CO 3	Apply stability criteria to evaluate the behavior of columns and beams under different loading conditions	**Apply (BT3)**
CO 4	Understand and explain the principles of failure theories and their applications in the design of mechanical components	**Understand (BT2)**

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Periods**
I.	**Stress, Strain, and Axial Loading**	Concept of stress and strain, normal and shear stress, stress tensor. Hooke's Law, Poisson's ratio, and elastic constants. Stress-strain diagrams for ductile and brittle materials. Thermal stresses and strains. Deformation of axially loaded members. Statically determinate and indeterminate problems. Compatibility conditions and	15
II.	**Bending and Torsion**	Shear force and bending moment diagrams. Pure bending, bending stresses, and section modulus. Shear stress distribution in beams. Deflection of beams: Double integration method, Macaulay's method, and moment-area method. Torsion of circular shafts, shear stress, and angle of twist. Power transmission by shafts.	15
III.	Energy Methods and Stability	Strain energy due to axial load, bending, and torsion. Castigliano's theorems and their applications. Principle of virtual work and its applications. Euler's theory of buckling for columns. Effective length and slenderness ratio. Rankine's formula and empirical methods.	15

IV.	Failure Theories and Design	Maximum principal stress theory, maximum shear stress theory, and distortion energy theory. Factor of safety and design considerations. Applications of failure theories in mechanical component design.	15
		TOTAL	60

Textbooks and References:

Textbook:

- R.C. Hibbeler, Mechanics of Materials, Pearson Education.
- Ferdinand P. Beer, E. Russell Johnston, Mechanics of Materials, McGraw-Hill.

Reference Books:

- Timoshenko and Gere, Mechanics of Materials, Cengage Learning.
- James M. Gere, Mechanics of Materials, Brooks/Cole.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 15 NCH = 60 NCH		**8 * 4 NCH = 32 NCH (Problem Solving, Assignments, Quiz, Presentation, Case Study, Discussion)**

Alternative NPTEL/SWAYAM Course:

S. No.	NPTEL Course Name	Instructor	Host Institute
1	Mechanics of Material	Prof. Priyanka Ghosh	IIT Kanpur

Online Link(s)
https://archive.nptel.ac.in/courses/105/106/105106172

Paper 111/Subject Name: Mathematics-III		Subject Code: MAT022C304
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: T

Objective:

This course aims to provide a strong foundation in probability theory and random variables, equipping students with the ability to analyze uncertainty and statistical relationships. It covers key concepts such as estimation theory, correlation, regression, and hypothesis testing for data-driven decision-making. Additionally, students will develop numerical methods to solve algebraic and transcendental equations while learning approximation techniques to enhance computational problem-solving skills.

Prerequisites: Sets and elements of Sets, Operation on Sets, Algebra of Sets.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**understand** probability, random variables, and statistical distributions	Understand (BT2)
CO 2	**apply** statistical methods like correlation, regression, and hypothesis testing .	Solve (BT3)
CO 3	**solve** equations using numerical methods like interpolation and integration	Apply (BT3)
CO 4	**analyze** the accuracy and efficiency of numerical techniques	Analyze (BT4)

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Probability**	Probability spaces, conditional probability, independence; Discrete random variables, Independent random variables, the multinomial distribution, Poisson approximation to the binomial distribution, infinite sequences of Bernoulli trials, sums of independent random variables; Expectation of Discrete Random Variables, Moments, Variance of a sum, Correlation coefficient, Chebyshev's Inequality. Continuous random variables and their properties, distribution functions and densities, normal, exponential and gamma densities. Bivariate distributions and their properties, distribution of sums and quotients, conditional densities, Bayes' rule.	**10**

II.	**Statistics**	Basic Statistics, Measures of Central tendency: Moments, skewness and Kurtosis -Probability distributions: Binomial, Poisson and Normal - evaluation of statistical parametersfor these three distributions, Correlation and regression – Rank correlation. Curve fitting bythe method of least squares- fitting of straight lines, second degree parabolas and more general curves. Test of significance: Large sample test for single proportion, difference of proportions, Tests for single mean, difference of means, and difference of standard deviations. Test for ratio of variances - Chi-square test for goodness of fitand independence of attributes	**10**
III.	**Numerical Methods-I:**	Finite differences, Relation between operators, Interpolation using Newton's forward and backward difference formulae.Interpolation with unequal intervals: Newton's divided difference and Lagrange's formulae. Numerical Differentiation, Numerical integration: Trapezoidal rule and Simpson's 1/3rd and 3/8 rules.	**15**
IV	**Numerical Methods-II:**	Solution of polynomial and transcendental equations – Bisection method, Newton-Raphson method and Regula-Falsi method. Numerical solution of ordinary differential equations: Taylor's series, Euler's methods. Runge-Kutta method of fourth order for solving first and second order equations. Predictor-corrector methods.	**10**
TOTAL			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH		8 * 3 NCH = 24 NCH (Problem Solving, Assignments, Quiz, Presentation, Case Study, Discussion)

Text Books:

1. Hoel P. G., Port S. C. and Stone C. J., *Introduction to Probability Theory*, 2003, Universal BookStall.
2. Satry S. S, *Introductory Methods of Numerical Analysis,* 4th edition, 2005, PHI.

Reference Books:

1. Grewal B. S., *Higher Engineering Mathematics*, 43rd edition 2014, Khanna Publishers.
2. Bali N. P. and Narayan Iyenger N *,A text book of Engineering Mathematics*, 9th edition, 2016, Laxmi Publication.
3. Kreyszig E. *Advanced Engineering Mathematics,* 9th edition, 2011, Wiley Eastern Ltd

Paper IV / Subject Name: Machine Drawing Lab		Subject Code: MEE022C319
L-T-P-C –1-0-4-3	Credit Units: 03	Scheme of Evaluation: P

Objective:

The objectives of the course are to teach the students to develop hands on skill in technical drawing required for manufacture and assembly of machine components.

Prerequisites: Engineering Graphics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	To understand national and international standards while drawing machine component	**BT 2**
CO 2	To familiarize in drawing assembly, orthographic and sectional views of various machine components	**BT 3**

Detailed Syllabus:

Submission of one drawing sheet for each of the following topics

1. **Introduction:** Classification of Machine Drawings, Principles of Drawings, Sectioning, Dimensioning, Limits, Fits and Tolerance, Symbols and Conventional Representation
2. **Orthographic Projections**: First angle and third angle projection, conversion of isometric views to orthographic views, sectional views
3. **Screwed Fasteners**: Thread nomenclature, forms of thread, designation, representation of threads, fasteners, bolts and nuts, locknuts
4. **Keys, Cotters and Pin joints**: Types of keys, Cotter joint and Knuckle joint
5. **Shaft Coupling**: Introduction, Rigid coupling and Flexible coupling
6. **Riveted Joints**: Introduction, rivets and riveting, types of rivet heads, types of riveted joints
7. **Assembly drawings**: Introduction, machine components, stuffing box, plumber block, screw jack
8. **Assembly drawings**: Introduction to engine parts

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
1 * 15 NCH = 15 NCH	4 * 15 NCH = 60 NCH	4 * 8 NCH = 32 NCH

Textbooks:

1. N.D. Bhatt, "*Machine Drawing*", Charotar Publishing House, 2008.

Reference Books:

1. Basudeb Bhattacharya, *Machine Drawing*, Oxford University Press
2. K.L. Narayana, "*Machine Drawing*", 3rd ed., New Age International, 200

Paper V/ Subject Name: Basic Electronics Engineering		Code: ECE022C306
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objective:

This course aims to provide fundamental knowledge of semiconductor devices, operational amplifiers, digital electronics, and communication systems. It covers the principles, applications, and circuit design of diodes, transistors, op-amps, logic gates, and modulation techniques, preparing students for practical implementations in electronic systems.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**recall** fundamental concepts of semiconductor devices, digital electronics, and communication systems	**BT 1**
CO 2	**understand** the working principles of semiconductor devices, including diodes, transistors, and voltage regulators	**BT 2**
CO3	**apply** Boolean algebra, logic circuits, and microprocessors/microcontrollers in digital system design.	**BT 3**
CO4	**analyze** electronic circuits, modulation techniques, and transmission systems for communication applications.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Semiconductor Devices and Applications**	Introduction to P-N junction Diode and V-I characteristics, Half wave and Full-wave rectifiers, capacitor filter. Zener diode and its characteristics, Zener diode as voltage regulator. Regulated power supply IC based on 78XX and 79XX series, Introduction to BJT, its input-output and transfer characteristics, BJT as a single stage CE amplifier,frequency response and bandwidth.	**15**
II.	**Operational amplifier and its applications**	Introduction to operational amplifiers, Op-amp input modes and parameters, Op- amp in open loop configuration, op-amp with negative feedback, study of practical op-amp IC 741, inverting and non-inverting amplifier applications: summing and difference amplifier, unity gain buffer, comparator, integrator and differentiator. **Timing Circuits and Oscillators**: RC-timing circuits, IC 555 and its applications as astable and mono-stable multi-vibrators, positive feedback, Barkhausen's criteria for oscillation, R-C phase shift and Wein bridge oscillator.	**15**

III.	**Digital Electronics Fundamentals**	Difference between analog and digital signals, Boolean algebra, Basic and Universal Gates, Symbols, Truth tables, logic expressions, Logic simplification using K-map, Logic ICs, half and full adder/subtractor, multiplexers, demultiplexers, flip-flops, shift registers, counters, Block diagram of microprocessor/microcontroller and their applications.	**15**
IV	**Electronic Communication Systems**	The elements of communication system, IEEE frequency spectrum, Transmission media: wired and wireless, need of modulation, AM and FM modulation schemes, Mobile communication systems: cellular concept and block diagram of GSM system.	**15**
TOTAL			**60**

Text /Reference Books:

1. Floyd ," *Electronic Devices"* Pearson Education 9th edition, 2012.
2. R.P. Jain , "*Modern Digital Electronics*" , Tata Mc Graw Hill, 3rd Edition, 2007.
3. Frenzel, " *Communication Electronics: Principles and Applications*" , Tata Mc Graw Hill, 3rdEdition, 2001

Course: B.Tech. (M.E)	**SYLLABUS**	**Semester: III**
Paper VI / Subject Name: Material Testing Lab		**Subject Code: 8**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Course Objective	*Teaching Learning Process*	*Learning Outcomes*	*Course Evaluation*
The objectives of the course are:- • To understand the procedure of doing different tests like hardness, compression, torsion, tension and impact etc in various materials	1. Demonstration 2. Lab Experiment 3. Quiz 4. Viva-Voce	On successful completion the student will be able to: • Describe the behavior of materials upon normal external loads. • Predict the behavior of the material under impact conditions. • Recognize the mechanical behavior of materials.	• Continuous Evaluation: 25% (Skill Test, lab copy, viva, lab involvement: Any Three) • Attendance: 5% • End term examination: 70 %

EXPERIMENT NO.	AIM OF THE EXPERIMENT
1	Tensile test on mild steel using UTM
2	Compression test on a given specimen using UTM
3	Bending test of rod on UTM
4	Determination of ultimate shear strength of mild steel
5	Torsion test on mild steel to determine modulus of rigidity
6	Determination of Impact strength using Izod impact testing machine
7	Determination of Impact strength using Charpy impact testing machine
8	Determination of hardness of steel by using Rockwell Hardness Test
9	Determination of stiffness and modulus of rigidity of Helical Spring
10	Determination of fatigue behavior of material by using Fatigue Testing Machine

DETAILED SYLLABUS OF 4th SEMESTER

Paper I/Subject Name: Fluid Mechanics		**Subject Code: MEE022C403**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

To equip students with the fundamental principles of fluid mechanics and hydraulic machines, enabling them to analyze, design, and optimize fluid systems and machinery for real-world engineering applications.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Explain the fundamental properties of fluids and their behavior under static and dynamic conditions.	Understand (BT2)
CO 2	Analyze fluid flow problems using principles of continuity, momentum, and energy	Solve
CO 3	Evaluate the performance of hydraulic machines such as pumps and turbines.	Apply (BT3)
CO 4	Demonstrate the use of computational tools for fluid flow analysis.	(BT Level: Apply)

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Periods**
I.	Fundamentals of Fluid Mechanics	Introduction to Fluid Mechanics: Definition of fluid, fluid properties: density, viscosity, surface tension, compressibility. Newtonian and non-Newtonian fluids. Fluid Statics: Pressure variation in fluids, hydrostatic forces on surfaces, buoyancy, and stability. Applications of fluid statics in engineering (e.g.,	15
II.	Fluid Dynamics and Flow Analysis	Fluid Dynamics: Continuity equation, momentum equation, and energy equation. Bernoulli's equation and its applications. Flow Measurement:Venturimeter, Orificemeter, Pitot tube. Dimensional Analysis and Similitude: Buckingham π-theorem, dimensionless numbers (Reynolds, Froude, Weber, Mach). Similitude and model studies.Flow Through Pipes: Laminar and turbulent flow, Darcy-Weisbach equation, friction factor, Moody's chart.Minor	15

		losses in pipes, pipe network analysis (series and parallel connections).	
III.	Hydraulic Machines: Pumps	Introduction to Hydraulic Machines: Classification and working principles.Pumps:Centrifugal pumps: working principles, performance characteristics, efficiency. Reciprocating pumps: working principles, performance characteristics. Cavitation and its prevention. Selection and design considerations for pumps.	15
IV.	Hydraulic Machines: Turbines and Computational Fluid Dynamics (CFD)	Turbines: Classification: impulse (Pelton wheel) and reaction (Francis, Kaplan) turbines. Working principles, performance characteristics, efficiency. Specific speed, unit quantities, and design considerations. Applications in hydroelectric power plants. Computational Fluid Dynamics (CFD): Introduction to CFD and its applications. Governing equations of fluid flow. Overview of CFD software tools (e.g., ANSYS, COMSOL)	15
		TOTAL	60

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 15 NCH = 60 NCH		**8 * 4 NCH = 32 NCH (Problem Solving, Assignments, Quiz, Presentation, Case Study, Discussion)**

Textbooks and References:

1. **Textbooks**:
    - Fluid Mechanics by R.K. Bansal.
    - Hydraulic Machines by R.K. Rajput.
    - Introduction to Fluid Mechanics by Fox and McDonald.
2. **References**:
    - Fluid Mechanics and Hydraulic Machines by Modi and Seth.
    - Computational Fluid Dynamics by John D. Anderson.

Alternative NPTEL/SWAYAM Course:

S. No.	NPTEL Course Name	Instructor	Host Institute
1	Fluid Mechanics	Prof. S.K. Som	IIT Kharagpur

Online Link(s)
https://nptel.ac.in/courses/112105129

Course: B.Tech. (M.E.)	SYLLABUS	Semester: IV
Paper II / Subject Name: Manufacturing Process I		Code: MEE022C401
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objective:

The objective of this course is to provide students with a fundamental understanding of casting, welding, bulk forming, and machining processes. It covers sand casting and other casting methods, gas and arc welding, advanced welding techniques, bulk deformation processes, and the working principles of lathe, shaper, and milling machines**,** equipping students with essential manufacturing knowledge.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**identify** fundamental manufacturing processes, including casting, welding, forming, and machining	**BT 1**
CO 2	**explain** the working principles and applications of various casting, welding, and machining techniques	**BT 2**
CO3	**demonstrate** the selection and application of appropriate manufacturing processes for different materials and products.	**BT 3**
CO4	**evaluate** the efficiency, advantages, and limitations of different manufacturing techniques in industrial applications	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**CASTING PROCESS**	Manufacturing; Definition and broad grouping Casting: Definition, Casting Materials Sand mould casting: Moulding sands - composition, properties & testing; Type of patterning; Design of gating system - sprue, runner, ingate & riser; Solidification – centre-line freezing Furnaces: Copula, Resistance furnace, Induction & Arc furnace. Casting Method: Centrifugal casting, Shell mould, Investment casting, Permanent mould casting, Die casting, and Slush casting. Casting defects, types, causes & remedy	**15**
II.	**WELDING PROCESS**	Major grouping of joining processes: welding, brazing and soldering. Broad classification of welding processes, types and principles Gas welding &thermit welding, Arc welding, Submerged arc welding, TIG & MIG, Plasma arc welding,	**15**

		Spot & butt welding, Hot forge welding, Friction welding, Pressure & percussion welding Ultrasonic welding, Laser beam welding, Electron beam welding Welding defects, types, causes & remedy	
III.	**FORMING**	**Metal Forming:** Hot and cold working of metals, classification of formingprocesses Rolling: Pressure and forces, Types of rolling mills, rolling defects **Forging**: Smith forging, Drop forging, Press forging, Machine forging,Forging defects **Extrusion**: Direct, Indirect, Impact, extrusion of tubes Sheet metal working: Bending, shearing, blanking & punching.	**15**
IV	**MACHINING PROCESSES**	Lathe – Functions, Classification & operations; Drilling Machine – Functions, Classification & operations Milling Machine – Functions, Classification & operations Finishing Processes	**15**
TOTAL			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH		8 * 3 NCH = 24 NCH (Problem Solving, Assignments, Quiz, Presentation, Case Study, Discussion)

Text Books:

1. P.N Rao, *Manufacturing technology, Foundry, Forming & Welding,* Tata McGraw-Hill Education,1992
2. A Ghosh & A Mullick, *Manufacturing Science*, 2nd Edition, 2010, East West Press
3. Choudhury H S K, *Elements Of Workshop Technology Vol-1(Manufacturing Processes)*, MediaPromoters, 2008

Reference Books:

1. R K Jain. *Production Technology: Manufacturing Processes, Technology and Automation,* 17thedition, 2004, Khanna Publishers
2. E.P Degarmo, Black & Kohser, *Materials & processes in manufacturing*, 11th edition, 2013,Wiley
3. S.K Sharma & S Sharma, *Manufacturing processes*, I.K International, 2013
4. S Kalpakjian; *Manufacturing Engineering & Technology*, 4th edition, Pearson Education India.

Alternative NPTEL/SWAYAM Course:

S.No.	NPTEL Course Name	Instructor	Host Institute
1	FUNDAMENTALS OF MANUFACTURING PROCESSESS	PROF. D.K DWIVEDI	IIT ROORKEE

Course: B.Tech. (M.E.)	SYLLABUS	Semester: IV
Paper III / Subject Name: Applied Thermodynamics		**Subject Code: MEE022C405**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the students to apply the concepts of thermodynamics to understand and analyze thermal devices

Prerequisites: Basic Thermodynamics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand various power cycles and working principles of various thermal devices	**BT 1 & BT 2**
CO 2	Utilize energy balance in thermal devices	**BT 3**
CO 3	Analyze energy conversion in various thermal devices	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Hours**
I	Power cycles	Vapor power cycles: Rankine cycle with superheat, reheat and regeneration. Gas power cycles, Air standard Otto, Diesel and Dual cycles, Air standard Brayton cycle: effect of reheat, regeneration and intercooling.	15
II	Boiler and Combustion	Classification of boilers, Mountings, Accessories, Evaporation capacity, Equivalent evaporation, Boiler efficiency, Heat balance sheet. Fuel & Combustion: Stoichiometry, air requirement, A/F ratio, heating value, Exhaust gas analysis.	15
III	Nozzle and Steam Turbine	Flow of steam through nozzles, critical pressure ratio, maximum mass flow rate through nozzle, effect of friction, nozzle efficiency. Impulse and reaction turbine, velocity diagram, work done and efficiencies.	15
IV	Compressor and Condenser	Reciprocating compressor, Effect of clearance, Volumetric efficiency, Compression ratio, Methods of improving Thermal efficiency, Compressor work, Multi-stage Compression, Intercooler and Aftercooler Classification of condensers, air leakage, condenser performance parameters	15
		TOTAL	**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 15 NCH = 60 NCH		4 * 8 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Project)

Textbooks:

1. P K Nag, *Engineering thermodynamics*, 6th edition, 2017, McGraw Hill Education.
2. Cengel and Boles, *Fundamentals of Thermodynamics*, 4th edition, 2001, Tata McGraw Hill Publication

Reference Books:

1. Sonntag, R. E, Borgnakke, C. and Van Wylen, G. J., 2003, 6th Edition, *Fundamentals of Thermodynamics*, John Wiley and Sons.
2. Jones, J. B. and Duggan, R. E., 1996, *Engineering Thermodynamics*, Prentice-Hall of India
3. Moran, M. J. and Shapiro, H. N., 1999, *Fundamental of Engineering Thermodynamic*, John Wiley and Sons.

Alternative NPTEL/SWAYAM Course:

S. No.	NPTEL Course Name	Instructor	Host Institute
1	Applied Thermodynamics	Prof. Niranjan Sahoo Dr. Pranab K Mondal	IIT Guwahati
Online Link: https://archive.nptel.ac.in/courses/112/103/112103307/			

Paper IV/Subject Name: Kinematics of Machines		Subject Code: MEE022C404
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: TP

Objective:

The objective of this course is to provide knowledge of transfer of motions and conversion of motions using mechanisms.

Prerequisites: A course on Engineering Mechanics, Statics and Dynamics.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the basic structure and elements of machines.	Understand (BT2)
CO 2	Determine position, velocity and acceleration of linkages in mechanism at any instant.	Solve (BT4)
CO 3	**Apply** the concepts of power transfer through belts, chains and gears to **analyze** load, speed, efficiency, and maintenance for **selecting** an effective transmission mechanism.	Apply (BT3)
CO 4	**Understand** basics related to friction and its practical application in mechanical engineering	Understand (BT2)

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	Introduction: Aims & scope of the course & Basic concepts of Mechanisms	Basic definitions, Difference between structure & Machine, Links & their types, Types of constrained motion, Kinematic pair & their classification, Grubler's mobility criteria, Inversion of a kinematic chain and applications, Hooks joint, Devis and Ackermann steering mechanism. An introduction to approximate and exact straight-line mechanism.	15
II.	Velocity and acceleration Analysis	Graphical (vector) method for velocity and acceleration of various mechanisms e.g. slider crank and four bar, Coriolis acceleration. Instantaneous Centre method, Kennedy's theorem, Klien's construction	15
III.	Transmission drives	Belt and Chain drives: Types and materials, Fundamentals of Power transmission, Phenomena of slip & creep, centrifugal and initial tensions, Tight side and slack side tensions, Conditions of max Power transmission. Classification of chain, chain length, angular speed ratio. Theory of gearing: Classification of gears and terminology, Law of gearing, gear profiles, Interference, and efficiency of gears, gear train, Torque analysis and various applications of complex gear trains.	15

IV.	Friction, Brakes and Clutches:	Introduction to friction, Law of friction, Coefficient of friction, Inclined Plane. Types of braking systems, force and torque analysis for block, band and band and block brake, disc brakes. Friction clutches: types, uniform pressure and uniform wear theory.	15
		TOTAL	60

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 15 NCH = 60 NCH		**8 * 4 NCH = 32 NCH (Problem Solving, Assignments, Quiz, Presentation, Case Study, Discussion)**

Textbooks:

1. Thomas Bevan, "Theory of Machines," CBS Publishers & Distributors, 2005.
2. W. L. Cleghorn, "Mechanisms of Machines," Oxford University Press, 2005.
3. R. L. Norton, "Kinematics and Dynamics of Machinery," Tata McGraw Hill, 2009.
4. A. Ghosh and A.K. Mallick, "Theory of Mechanisms and Machines," Affiliated East-West Pvt. Ltd, New Delhi, 1988.

References:

1. Rattan, S.S.,"Theory of Machines", Tata McGraw-Hill Publishing Company Ltd.,New Delhi, 2009.
2. Uicker, J.J., Pennock, G.R and Shigley, J.E., "Theory of Machines and Mechanisms", Oxford University Press,New Delhi, 2009
3.Khurmi, R.S., and Gupta, J.K., "Theory of Machines", S.Chand& Company, 2009.
4. Kinematics by HN Tyson
5. Rao, J.S., and Dukkipati, R.V, "Mechanism and Machine Theory", New Age International (P) Ltd Publishers. New Delhi, 2007.

Alternative NPTEL/SWAYAM Course:

S. No.	NPTEL Course Name	Instructor	Host Institute
1	Kinematics of Mechanisms and Machines	Prof. A. Dasgupta	IIT Kharagpur

Online Link(s)
https://nptel.ac.in/courses/112105268

Paper V/Subject Name: Engineering Materials and Applications		Subject Code: MEE022C402
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

- The objective of the course is to provide students with a comprehensive understanding of the structure, properties, selection, and applications of engineering materials, enabling them to analyze material behavior, explore advanced materials, and apply this knowledge to solve real-world engineering problems.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Classify and describe the structure, properties, and applications of various engineering materials. (BT Level: Understand)	Understand (BT2)
CO 2	Analyze the mechanical behavior of materials under different loading and environmental conditions. (BT Level: Analyze)	Solve (BT4)
CO 3	Select appropriate materials for specific engineering applications based on their properties. (BT Level: Apply)	Apply (BT3)
CO 4	Evaluate the role of advanced materials in modern engineering applications. (BT Level: Evaluate)	Understand (BT2)

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I.	**Introduction to Engineering Materials**	Classification of engineering materials: Metals, Polymers, Ceramics, Composites, and Advanced Materials. Atomic structure and bonding in materials. Crystal structures: BCC, FCC, HCP, and defects in crystals. Mechanical properties of materials: Stress-strain curves, hardness, toughness, fatigue, and creep.	15
II.	**Ferrous and Non-Ferrous Materials**	Ferrous materials: Cast iron, plain carbon steels, alloy steels, and stainless steels. Heat treatment of steels: Annealing, normalizing, hardening, tempering, and case hardening. Non-ferrous materials: Aluminum, copper, titanium, and their alloys. Applications of ferrous and non-ferrous materials in engineering.	15

III.	**Non-Metallic and Composite Materials**	Polymers: Types, properties, and applications. Ceramics: Structure, properties, and applications. Composite materials: Classification, properties, and applications. Fiber-reinforced composites: Types and manufacturing processes. Material selection for specific engineering applications	15
IV.	**Advanced Materials and Applications**	Smart materials: Shape memory alloys, piezoelectric materials, and their applications. Nanomaterials: Properties, synthesis, and applications. Biomaterials: Types and applications in medical engineering. Emerging materials: Graphene, carbon nanotubes, and their applications. Sustainability in material selection: Eco-friendly materials and recycling.	15
		TOTAL	60

1. **Textbooks and References:**
    1. **Textbooks:**
        - Callister, W. D., & Rethwisch, D. G. *Materials Science and Engineering: An Introduction*. Wiley.
        - Smith, W. F., & Hashemi, J. *Foundations of Materials Science and Engineering*. McGraw-Hill.
        - Raghavan, V. *Materials Science and Engineering: A First Course*. PHI Learning.
    2. **References:**
        - Ashby, M. F., & Jones, D. R. H. *Engineering Materials 1: An Introduction to Properties, Applications, and Design*. Elsevier.
        - Budinski, K. G., & Budinski, M. K. Engineering Materials: Properties and Selection. Pearson.
        - Askeland, D. R., & Fulay, P. P. Essentials of Materials Science and Engineering. Cengage Learning.

Alternative NPTEL/SWAYAM Course:

S. No.	NPTEL Course Name	Instructor	Host Institute
1	Introduction to Materials Science and Engineering https://nptel.ac.in/courses/112105268	Prof. Satish V. Kailas,	IISc Bangalore

Paper VI/Subject Name: Fluid Mechanics Lab		Subject Code: MEE022C411
L-T-P-C – 0-0-2-1	Credit Units: 01	Scheme of Evaluation: P

Objective:

To equip students with the fundamental principles of fluid mechanics and hydraulic machines, enabling them to analyze, design, and optimize fluid systems and machinery for real-world engineering applications.

Course Outcomes:

On successful completion of the course the students will be able to:	
SI No	**Course Outcome**
CO 1	Demonstrate the ability to measure fluid properties such as viscosity, density, and pressure.
CO 2	Analyze and interpret data from experiments related to fluid flow and hydraulic machines.
CO 3	**Evaluate the performance of hydraulic machines such as pumps and turbines.**
CO 4	Operate and evaluate the performance of hydraulic machines such as pumps and turbines.

List of Experiments:

The following experiments are designed to cover the fundamental and applied aspects of fluid mechanics:

1. **Bernoulli's Theorem Verification:** Experimental verification of Bernoulli's equation using a Bernoulli's apparatus.
2. **Performance Characteristics of Centrifugal Pump:** Determination of head, discharge, and efficiency of a centrifugal pump.
3. **Performance Characteristics of Pelton Wheel Turbine:** Determination of efficiency and power output of a Pelton wheel turbine.
4. **Performance Characteristics of Francis Turbine:** Determination of efficiency and power output of a Francis turbine.
5. **Performance Characteristics of Kaplan Turbine:** Determination of efficiency and power output of a Kaplan turbine.

Textbooks:

- Fluid Mechanics and Hydraulic Machines by R.K. Rajput
- Introduction to Fluid Mechanics by Fox and McDonald

Online Resources:

- NPTEL lectures on Fluid Mechanics.
- AICTE-approved e-learning platforms.
- **Laboratory Manuals:**

Paper VII/Subject Name: Kinematics of Machine Lab		**Subject Code: MEE022C414**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

List of Experiments:

1. To draw velocity diagram of four bar mechanism.
2. To draw velocity diagram of slider crank mechanism.
3. To draw acceleration diagram of four bar mechanism.
4. Study of Ackerman's Steering Gear Mechanism.
5. To study various types of gears – Helical, cross helical worm, bevel gear.
6. To study various types of gear trains – simple, compound, reverted, epicyclic and differential.

DETAILED SYLLABUS OF 5th SEMESTER

Paper I / Subject Name: Machine Element & System Design		**Code: MEE022C501**
L-T-P-C –4-0-0-4	**Credit Units: 4**	**Scheme of Evaluation: T**

Objectives:

1. To understand safety-critical design of machine components using failure criteria based on mechanics of materials
2. To understand the origins, nature and applicability of empirical design principles, relevant codes, standards and design guidelines for different machine elements
3. To appreciate the relationships between component level design and overall machine system design and performance

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**BloomsTaxonomy Level**
CO 1	Recall and explain the fundamental principles of machine elements and their roles in mechanical systems..	**BT 1**
CO 2	Describe and interpret the causes and mechanisms of failure in various machine elements.	**BT 2**
CO3	Apply appropriate design codes, standards, and guidelines for the selection and evaluation of machine components.	**BT 3**
CO4	Analyze the performance and interaction of machine elements in mechanical system assemblies.	**BT 4**

Detailed syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I.	**Introduction to Machine Elements and System Fundamentals**	Anatomy of machines; Functional dissection of motorcycle, washing machine, sewing machine, etc. into machine elements including gears, rack and pinions, cams, chains, belts, pulleys, flywheels, bearings, shafts, keys, brakes, etc.; Design considerations – Limits, fits and standardization; Friction and lubrication.	**15**
II.	**Force Analysis and Failure Theories**	Force analysis of machine elements and machine systems; Application to power screws and couplings, clutches, and brakes. Static failure theories including normal stress theory, shear stress theory, distortion energy theory; von Mises stress; Factor of safety; Stress concentration factors; Fatigue failure theories: mean and alternating stresses, yield,	**15**

		ultimate, and endurance strength; Goodman, Gerber, and Soderberg lines.	

Modules	Topics	Course Contents	Hours
III.	**Design of Machine Elements**	Springs – Helical compression, tension, torsional and leaf springs; Fasteners – threaded fasteners, bolted joints, preloaded bolts, rivets and welded joints; Shafts – shafts under static and fatigue loadings; Keys; Sliding and rolling contact bearings; Transmission elements – transmission ratio and efficiency of spur, helical, bevel and worm gears; belt and chain drives; Flywheels.	**15**
IV.	**Dynamics and Application of Mechanical Systems**	Single degree-of-freedom systems; Natural frequency and critical damping; Forced vibration; Resonance; Balancing of reciprocating and rotating masses; Torsional vibration and critical speeds of shafts. Case studies on automobile suspensions, automatic transmissions, material conveyor systems, construction machinery, etc.	**15**
TOTAL			**60**

Text /Reference Books:

1. Shigley, J.E. and Mischke, C.R., "Mechanical Engineering Design," McGraw-Hill, 1989.
2. Deutschman, D., & Wilson, C.E., "Machine Design Theory & Practice," Macmillan, 1992
3. Juvinal, R.C., "Fundamentals of Machine Component Design," John Wiley, 1994.
4. Spottes, M.F., "Design of Machine elements," Prentice-Hall India, 1994
5. R. L. Norton, "Mechanical Design – An Integrated Approach," Prentice Hall, 2009
6. Sadhu Singh, "Machine Design", Khanna Book Publishing, 2021.

Online Resources:

1 https://archive.nptel.ac.in/courses/112/105/112105124/

Paper II / Subject Name: Manufacturing Process II		Code: MEE022C502
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objectives:

To motivate and challenge students to understand and develop an appreciation of the processes in correlation with material properties which change the shape, size and form of the raw materials into the desirable product by conventional or unconventional manufacturing methods

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and describe the principles of various conventional and unconventional manufacturing processes.	**BT 1**
CO 2	Explain the selection criteria of suitable manufacturing methods for different materials and product types.	**BT 2**
CO3	Apply process parameters to plan basic operations in casting, forming, welding, and machining.	**BT 3**
CO4	Analyze and compare different manufacturing processes for their efficiency, material compatibility, and economic viability.	**BT 4**

Detailed syllabus:

Modules	Topics	Course Contents	Hours
I.	**Fundamentals of Manufacturing and Process Classification**	Additive, subtractive and shaping processes; Relative advantages and limitations; Inter-dependency of geometry, material and process; Effect on product quality and cost; Part design for manufacturability; Process selection criteria	**15**
II.	**Shaping and Joining of Materials**	Metal casting (sand, die and investment casting), Bulk forming (forging, rolling, extrusion, drawing) and sheet forming (shearing, deep drawing, bending); Thermoplastic and thermoset plastic processes (ex. injection and blow molding); Powder metallurgy; Metal injection molding; Glass and composite processes (layup)	**15**
III.	**Material Removal and Surface Processes**	Turning, Drilling, Milling, Grinding and other finishing processes; Single and multi-point cutting tools; Cutting tool materials; Cutting fluids; Material removal rates, surface finish, accuracy, integrity and machinability	**15**

IV.	**Advanced, Additive, and Process Modeling Techniques**	Abrasive Jet Machining, Water Jet Machining; Ultrasonic Machining; Electrical Discharge Machining, Wire EDM; Electro Chemical Machining; Laser Beam Machining, Plasma Arc Machining and Electron Beam Machining; Micro and nano manufacturing. **Additive Manufacturing Processes:** Extrusion; vat polymerization, powder bed fusion; material jetting, binder jetting; direct energy deposition and lamination processes . **Joining and Fastening Processes:** Arc welding, gas welding, shielded metal arc welding; GMAW (MIG) and GTAW (TIG); Brazing and soldering; Solid state joining; Adhesive bonding. **Manufacturing Process Modeling** (for any one process, including simulation and industrial case study): Casting – metal flow, solidification and cooling; application to design of gating and feeding systems for quality and yield optimization; OR Forming – Plastic deformation and yield criteria; load estimation; OR Machining – Orthogonal cutting, various force components; Chip formation, Tool wear and tool life.	**15**
TOTAL			**60**

Text /Reference Books:

1. Amitabha Ghosh and A.K. Mallick, Manufacturing Science. Affiliated East-West Press Pvt. Ltd. 2010.
2. Kalpakjian and Schmid, Manufacturing Processes for Engineering Materials, Pearson India, 2014
3. M. P. Groover, Fundamentals of Modern Manufacturing: Materials, Processes, and Systems 4. Degarmo, Black & Kohser, Materials and Processes in Manufacturing.

Online Resource:

https://www.mooc-list.com/tags/manufacturing

Paper III / Subject Name: Metrology and Measurements		Code: MEE022C503
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objective:

The objectives of this course are to:

- Introduce fundamental principles of engineering metrology and measurement.
- Develop understanding of measurement systems, standards, and calibration.
- Familiarize students with linear, angular, and form measurement techniques.
- Impart knowledge of limits, fits, tolerances, and surface metrology.
- Enable interpretation of measurement data with statistical quality control concepts.

Course Outcomes:

On successful completion of the course, the students will be able to:		
Sl. No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Explain the concepts of measurement, standards, errors, and calibration used in engineering metrology.	**Understand (BT2)**
CO2	Apply limits, fits, and tolerances for functional and interchangeable design.	**Apply (BT3)**
CO3	Analyze dimensional, geometric, and surface measurements using appropriate metrology instruments.	**Apply (BT3) & Analyze (BT4)**
CO4	Evaluate advanced and automated measurement systems for quality assurance in manufacturing.	**Analyze (BT4)**

Detailed Syllabus:

Modules	**Topic**	**Course Content**	**Periods**
I	Fundamentals of Metrology and Measurement & Limits, Fits, Tolerances, and Geometric Measurement	Parameters – geometry (straightness, flatness, roundness, etc.); Definitions: Accuracy, precision, range, resolution. Structure and examples of measurement systems; Calibration principles; Linear and angular measurements; Comparators; Gauge design. Definitions; Tolerance zone and grades, Hole and shaft system, Geometric tolerances, Tylor's principle of gauging; Tolerance analysis in manufacturing and assembly; GO NO-GO gauges	15
II	Measurement	Linear & Angular: Definition and principle: Vernier, Micrometre, Slip Gauges, Angle Gauges, Sine Bar, Spirit Level, Clinometer, Angle Dekkor, Autocollimator, Taper Measurement: Tapered Ring and Hole; Tapered Shaft and Plug Gauges; V-Block, Dovetail; Internal Diameter Measurement: Inside Micrometre, 2-ball & 3-ball methods; Pin method.	15

III	Screw Thread and Surface Measurement	Principle and Measurement of major, minor and effective diameter: Bench micrometre, 2-wire method, 3-wire method; Pitch Measurement: Screw pitch gauge; Thread Angle Measurement. Definitions: Roughness, Waviness, Defect or Flaw, Lay Direction; Measurement of Surface Roughness and Errors; Tomlinson Surface meter, Surface Profilograph, Abbot's Profilometer, Talysurf, Perthometer, PEC material, Plastic Replica Technique	15
IV	Quality Control	Introduction, types of inspection, Statistical Quality Control (SQC), Control Charts (Control charts for attributes, & control charts for variables), applications of control charts, Sample Inspection, Single & double sampling plans. Introduction to ISO: 900 and ISO: 14000; Basic concept of Total Quality Management (TQM), Quality Circle, and Quality Assurance	15
Total			60

Textbooks and References:

1. E.O Doebelin and Dhanesh Manik, "Measurement Systems", McGraw Hill, 2017

2. Bewoor & Kulkarni, "Metrology & Measurement" Tata McGraw Hill, 2009.

3. D. James, and S, Meadow, "Geometric Dimensioning and Tolerancing", Marcel Dekker, 1995

4. Madhav S. Phadke, Quality Engineering using Robust Design, Prentice Hall, 1989

Paper IV / Subject Name: Heat and Mass Transfer		Code: MEE022C504
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objectives:

1. Build a solid foundation in heat transfer, exposing students to the three basic modes namely conduction, convection and radiation.
2. Rigorous treatment of governing equations and solution procedures for the three modes, along with solution of practical problems using empirical correlations.
3. The course will also briefly cover boiling and condensation heat transfer, and the analysis and design of heat exchangers.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify and describe the fundamental principles of conduction, convection, and radiation.	**BT 1**
CO 2	Formulate and analyze steady and unsteady heat transfer problems involving one or more modes.	**BT 2**
CO3	Apply analytical, empirical, or approximate methods to evaluate temperature distributions and heat transfer rates.	**BT 3**
CO4	Design heat exchangers and determine insulation requirements to control thermal energy losses.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Conduction Heat Transfer**	Three modes of heat transfer; Examples of equipment (like air conditioner and air cooler) involving heat transfer; Derivation of heat balance equation. Steady 1D solution for conduction heat transfer in Cartesian, cylindrical and spherical geometry; Concept of conduction and film resistances; Critical insulation thickness; Lumped system approximation and Biot number; Heat transfer through pin fins; 2D conduction solutions for steady and unsteady heat transfer	**15**
II.	**Convection Heat Transfer**	Basic equations; Boundary layers; Forced convection; External and internal flows; Natural convective heat transfer; Dimensionless parameters for forced and free convection heat transfer; Correlations for forced and free convection; Approximate solutions to laminar boundary layer equations for internal and external flow; Estimating heat transfer rates in laminar and turbulent flow situations using appropriate correlations for free and forced convection	**15**

III.	**Radiation Heat Transfer**	Interaction of radiation with materials; Definitions of radiative properties; Stefan Boltzmann's law; Black and grey body radiation; Calculation of radiation heat transfer between surfaces using radiative properties; View factors and the radiosity method; Examples for two-body enclosures; Radiation shield.	**15**
IV.	**Heat Exchanger Design**	Function, classification and configuration of heat exchangers; Evaluation of mean temperature difference; Heat exchanger effectiveness; Analysis, design and selection of heat exchangers. **Boiling and Condensation heat transfer** Pool boiling; Flow boiling; Film and drop wise condensation	**15**
TOTAL			**60**

Text Books:

1. J.P. Holman and S. Bhattacharyya, "Heat Transfer," McGraw Hill, 2017.
2. F.P. Incropera, and D.P. Dewitt, "Fundamentals of Heat and Mass Transfer," John Wiley, 2019
3. Yunus A Cengel, "Heat Transfer: A Practical Approach," McGraw Hill, 2002.

Reference Books:

1. A. Bejan, "Heat Transfer," John Wiley, 1993

Online Resources:

1 https://onlinecourses.nptel.ac.in/noc22_ch65/preview

Paper V / Subject Name: Heat Transfer Lab		**Code: MEE022C514**
L-T-P-C –0-0-2-1	**Credit Units: 1**	**Scheme of Evaluation: P**

Course Objective

- To expose the students to the basic knowledge of heat transfer and help them to develop experimental skills.
- To study the concepts, applications of the heat transfer laboratory.

Detailed Syllabus:

EXPERIMENT NO.	AIM OF THE EXPERIMENT	HOURS
1	Heat transfer through composite wall	3
2	Emissivity measurement apparatus	3
3	Forced convection apparatus	3
4	Specific heat apparatus (Heat pipe)	3
5	Heat transfer co-efficient for vertical tube (Natural Convection)	3
6	Parallel flow / counter flow heat exchanger	3
7	Boiling heat transfer	3
8	Stefan Boltzmann apparatus	3

Course: B.Tech. (M.E)	**SYLLABUS**	**Semester: V**
Paper VI / Subject Name: Dynamics of Machines		**Subject Code: MEE022C505**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Course Objective	*Teaching Learning Process*	*Learning Outcomes*	*Course Evaluation*
The objectives of the course are:- • To provide fundamentals of the basic components of forces acting over the machine components. • To derive the equation motion of SDOF by using different methods such as energy method, Rayleigh's method and newton's method. • To provide understanding of the response of the system for various unbalances acting on the system • To evaluate the natural frequencies of multi DOF system.	5. Lecture 6. Assignment 7. Individual and Group Presentation 8. Quiz 9. Class test 10. Viva-Voce	On successful completion the student will be able to: • Demonstrate an understanding of turning moment diagrams in various applications. • Demonstrate skills to design flywheel for an IC engine and punching press with the consideration of geometrical and economic constraints. • Participate or lead projects in mechanical system design for the controlling of vibration.	• Continuous Evaluation: 15% (Assignment, Class Test, Viva, Seminar, Quiz : Any Three) • Mid-term examination: 10% • Attendance: 5% • End Term Examination: 70%

Detailed Syllabus:

Modules	Topics / Course content	Hours
I.	**Force Analysis** Applied and constraint forces – Free body diagrams – Static equilibrium conditions – Two, three & four members – Static force analysis of simple mechanisms – Dynamic force analysis –Inertia force and Inertia torque – D Alembert's principle – The principle of superposition –Dynamic Analysis in reciprocating engines – Gas forces – Equivalent masses – Bearing loads– Crank shaft torque – Turning moment diagrams – Fluctuation of energy – Fly Wheels –Engine shaking forces.	**15**

II.	**Free and Forced Vibration of SDOF** Basic features of vibratory systems – Degrees of freedom–Natural frequency Derivation of equation motion – newton's method, energy method, Rayleigh's method,– Damped free vibration,- Resonance– Magnification factor, Response of rotating and reciprocating unbalance, support motion, vibration isolation, transmissibility – Critical speeds of shafts – Torsional vibration – Torsionally equivalent shaft – Two and three rotor systems.	**15**
III.	**Two Degrees and Multi Degrees of Freedom System** Torsional vibration of two rotor system, vibrations of two DOF system, semi-definite system, Co-ordinate coupling, vibration absorber-Derivation of influence co-efficient matrix, generalized co-ordinates, Matrix method, Orthogonality principle, Matrix Iteration Method, Dunkerley's method, Holzer's Method,	**15**
IV.	**Dynamics of Governors and CAM** Governors – Types – Centrifugal governors – Gravity controlled and spring controlled centrifugal governors – Characteristics – Effect of friction – Controlling force –. Classification of cams and followers – Terminology and definitions – Displacement diagrams –Uniform velocity, parabolic, simple harmonic, cycloidal and polynomial motions **Balancing of Inertia forces and Gyroscopic Action** Balancing of rotating masses, two-plane balancing, determination of balancing masses, balancing of Internal Combustion Engines. Gyro dynamics, free motion of symmetrical gyroscope, gyroscopic moment of a symmetrical gyroscope in regular precession, gyroscopic effects in machines	**15**
Total		**60**

Text Books:

1. Amitabha Ghosh, Asok Kumar Mallik, "Theory of Mechanisms and Machines", East-West Press.
2. V.P.Singh, "Mechanical Vibrations", Dhanpat Rai & Co
3. Rao.J.S. and Dukkipati.R.V. 'Mechanisms and Machine Theory', Wiley-Eastern Ltd., New Delhi, 1992.

Reference Books:

1. S S Rattan, "Theory of Machines", McGraw Hill publications.

Paper VII / Subject Name: Dynamics of Machines Lab	Subject Code: MEE022C515	
L-T-P-C – 0-0-2-1	Credit Units: 01	Scheme of Evaluation: P

Objectives:

- To understand the force-motion relationship in components subjected to External Forces.
- To analyse the force-motion characteristics of standard mechanisms

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Analyse the behaviour of bodies undergoing vibrations, both theoretically as well as practically.	**BT 4**
CO 2	Analyse and interpret data across all practical.	**BT 4**

Detailed Syllabus:

Experiment No	Aim of the Experiment	Hours
1.	To Study the undamped free vibration of spring mass system	03
2.	To study the forced vibration of spring mass system	03
3.	To find the natural frequencies of a two rotor system.	03
4.	To study the whirling phenomenon of shaft.	03
5.	To study the cam analysis by graphical method	03
Total		**15**

Course: B.Tech. (M.E)	**SYLLABUS**	**Semester: V**
Paper VIII/ Subject Name: Metrology and Measurements Lab		**Subject Code: MEE022C513**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Course Objective	*Teaching Learning Process*	*Learning Outcomes*	*Course Evaluation*
The objectives of the course are: - • To provide students with the necessary skills for calibration and testing of different gauges and instruments. • To provide students with the necessary skills to collect data, perform analysis and interpret results to draw valid conclusions through standard test procedures using various metrology instruments.	11. Demonstration 12. Lab Experiment 13. Quiz 14. Viva-Voce	On successful completion the student will be able to: • Use common metrology instruments for dimensional measurement. • Perform basic calibration and measurement procedures. • Apply measurement techniques for quality control in engineering applications.	• Continuous Evaluation: 25% (Skill Test, lab copy, viva, lab involvement: Any Three) • Attendance: 5% • End term examination: 70 %

Detailed Syllabus:

Experiment No	**Aim of the Experiment**	**Hours**
1.	To calibrate the given Vernier Calliper using Slip Gauge	03
2.	To calibrate the given Micrometer using Slip Gauge	03
3.	To measure the external diameter of the given specimen using vernier calliper	03
4.	To obtain the thickness and diameter of the work piece using the micrometer	03
5.	To measure the height of given work piece by using vernier height gauge	03
6.	To measure the angle in the given work piece using Bevel Protractor	03
7.	To measure the taper angle of a given work piece by using Sine bar	03
8.	To measure the thickness and depth of the gear using gear tooth vernier	03
9.	To find Pitch Circle Diameter (P.C.D) of a screw	03
Total		**27**

DETAILED SYLLABUS OF 6th SEMESTER

Paper VI / Subject Name: Robotics & Control		**Code: MEE022C601**
L-T-P-C –4-0-0-4	**Credit Units: 4**	**Scheme of Evaluation: T**

Objectives:

1. Model and analyze mechatronic systems for an engineering application
2. Identify sensors, transducers and actuators to monitor and control a process or product.
3. Develop PLC programs for an engineering application.
4. Evaluate the performance of mechatronic systems.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify and recognize common electro-mechanical systems and their applications in daily life.	**BT 1**
CO 2	Explain the role and working of sensors, actuators, and controllers in integrated mechatronic systems.	**BT 2**
CO 3	Apply control theory principles to design basic feedback control systems for automation.	**BT 3**
CO 4	Analyze measurement systems for various physical quantities in terms of accuracy, range, and their use in automatic control	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I.	**Introduction to sensors and actuators**	Electro-mechanical systems; Typical applications; Examples – automobiles, home appliances, medical instruments, etc. Transduction principles; Sensitivity, accuracy, range, resolution, noise sources; Sensors for common engineering measurements – proximity, force, velocity, temperature, etc.; Signal processing and conditioning; Selection of sensors Pneumatic and hydraulic actuators; Electric motors including DC, AC, BLDC, servo and stepper motors; Solenoids and relays; Active materials – piezoelectric and shape memory alloys	**15**
II.	**Machine Controls**	Microprocessors and their architecture; Memory and peripheral interfacing; Programming; Microcontrollers; Programmable Logic Controllers; PLC principle and operation; Analog and digital input/output modules; Memory module; Timers, internal relays, counters and data handling; Industrial automation systems; Basic PLC programming; Industry kits (Arduino, Raspberry Pi, etc.).	**15**

III.	**Robotics**	Robot configurations: serial and parallel; Denavit–Hartenberg parameters; Manipulators kinematics; Rotation matrix, Homogenous transformation matrix; Direct and inverse Kinematics for robot position and orientation; Workspace estimation and path planning; Robot vision; Motion tracking; Robot programming and control; Industrial robots - Pick and place robots, sorting, assembly, welding, inspection, etc.	**15**
IV	**Control Theory and Systems**	Basic control concepts; Feedback; Open and closed loop control; Concept of block diagrams; P, PI and PID controllers; Tuning the gain of controllers; System models, transfer functions, system response, frequency response; Root Locus method and Bode plots **Computational Tools:** Demonstration and projects using simulation software (e.g., Matlab, Scilab, ROBODK) for control systems and robotics.	**15**
TOTAL			**60**

Text Books:

1. W. Bolton, "Mechatronics," Addison Wesley Longman, 2010.
2. J. J. Craig, Introduction to Robotics Mechanics and Control, Addison Wesley, 1999.

Reference books:

3. G.K. McMillan, "Process/Industrial Instruments and Controls Handbook," McGraw-Hill, 1999.
4. S. Mukherjee, "Essentials of Robotics Process Automation", Khanna Book Publishing, 2021.

Online Resources:

1. **https://nptel.ac.in/courses/107/106/107106090/**
2. **https://nptel.ac.in/courses/112/101/112101098/**
3. **https://nptel.ac.in/courses/112/107/112107289/**
4. **https://nptel.ac.in/courses/112/104/112104298/**

Paper II / Subject Name: IC Engine		Code: MEE022C602
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objectives:

- To develop the concept of heat engine design.
- To provide understanding of the working principle of designs of engine sub systems.
- To introduce and develop the concepts of engine performance evaluation and testing.
- To provide understanding knowledge of the cause, effect and control of emission from I C engine

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and explain the fundamental principles of internal combustion engines.	**BT 1**
CO 2	Describe and interpret the function and design of various IC engine subsystems (fuel, cooling, lubrication, etc.).	**BT 2**
CO 3	Apply thermodynamic principles to analyze combustion and engine cycles.	**BT 3**
CO 4	Evaluate the performance of an IC engine through testing and interpret key parameters like efficiency and power output.	**BT 4**

Detailed Syllabus:

Modules	Topics/	Course content	Hours
I.	Introduction and Classification of Heat Engines.	Engine components and working. Two-stroke, four-stroke. Air standard cycles, analysis of Otto, Diesel and Dual engines. Fuel-air cycles and actual cycles. Measurement of speed, torque, fuel consumption, determination of IHP, BHP and FHP, specific fuel consumption, determination of indicated thermal efficiency, brake thermal efficiency and mechanical efficiency, plot of efficiency vs. speed curves. Speed-torque curve with *bsfc* muscle curve, Morse Test.	**15**
II.	Classification of hydrocarbons in fuels,	Classification and desirable characteristics of I.C. engine fuels, Rating of S.I. and C.I. engine fuels, Laboratory method of determining Octane and Cetane Number. Alternative fuels (liquid, gaseous, etc.). Analysis of combustion product, HCV and LCV of the fuels. Combustion in I C Engines: Homogenous and heterogenous mixture, Stages of Combustion in S.I and C.I engines, Flame front propagation, Parameter influencing combustion, Detonation and knocking in S.I. and C.I. engines and their	**15**

		preventions, Combustion chamber design basics in SI Engine & CI Engines	
III.	Carburettor simple, Mechanical fuel injection system:	Analysis of simple carburettor, Mechanical fuel injection system: classification and working. Types of Injection pumps and nozzles. Mechanism of Spray formation, Calculation of fuel spray velocity. EFI system: working, sensors, MPFI, GDI Ignition systems in I.C. engines (Battery, magneto and electronic), ignition timing and spark advance.	**15**
IV.	Supercharging:	Types of superchargers, methods, effect and limitations of supercharging, Turbo charging, Downsizing and down-speeding of IC engines. Emission norms, Emission from IC Engines: classification (Exhaust and Non-exhaust), formation, control methods - catalytic and thermal converters, EURO and BS Norms Lubrication System: Function, types and working principle, properties of lubricating oil. Cooling system: Function, Types and working principle EMS (Engine Management System)	**15**
Total			**60**

Text Books:

1. V Ganesan, "Internal Combustion Engines", Mc Graw Hill Education (India) Pvt. Ltd., 4th Edition, 2012
2. M L Mathur, R. P. Sharma, "Internal Combustion Engines", Dhanpat Rai & Sons, 2014
3. H. N. Gupta, "Fundamentals of Internal Combustion Engines", PHI Learning Pvt. Ltd. University Press. 2nd edition, 2013
4. J. B. Heywood, "Internal Combustion Engine fundamentals", Mc Graw Hill Education, 1 edition, 2017

Paper III/ Subject Name: Computer Aided Design and Analysis		Code: MEE022C603
L-T-P-C –3-0-2-4	Credit Units: 4	Scheme of Evaluation: T

Objective:

To provide an overview of how computers can be utilized in mechanical component design

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the fundamentals of computer-aided design and engineering tools.	**BT 1**
CO 2	Explain the basic features of CAD software for modeling mechanical components.	**BT 2**
CO 3	. Use CAD tools to model and simulate simple mechanical components and assemblies.	**BT 3**
CO 4	Analyze the behavior of modeled components using basic CAE techniques (e.g., FEA or motion analysis).	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Introduction**	Role of computers in design process; Computer aided design, analysis and manufacturing; Computer integrated manufacturing; Popular CAD software used in industry; Input and output devices **Transformations:** Matrix representation of points, lines and planes; 2D transformation for translation, scaling, rotation and reflection; Homogeneous representation & concatenation; 3D transformations.	**15**
II.	**Curves and Surfaces,** **Solid modelling techniques**	Representation of curves; Hermite curves, Bezier curves, B-spline curves, Rational curves; Surface modelling – parametric representation, planar surface, surface of revolution, Coons and bicubic patches, Bezier and B-spline surfaces. **Solid modelling techniques** – sweep (linear and curved), Boolean (constructive solid geometry) and other techniques; Solid model representation (Boundary and Constructive Solid Geometry); Medical modelling (pixels, scans and voxels); Exchange standards (IGES, DXF, STEP, STL etc.).	**15**
III.	**Engineering Analysis**	Introduction to finite element method; Principle of potential energy; FE analysis of 1D element problems (spring, bar, truss elements); Development of element stiffness equation and their assembly; Plain strain and plain stress problems; Domain discretization, pre-processing and post-processing; Verification and validation; Popular CAE software used in industry	**15**

IV.	**Design Optimization**	Purpose and application of optimum design, Primary and subsidiary design equations, Limit Equations, Normal, redundant and incompatible specifications problems; Computer-aided design optimization.	**15**
TOTAL			**60**

Text Books:

1. Ibrahim Zeid, "Mastering CAD CAM," Tata McGraw Hill Publishing Co. 2007.
2. C. McMohan and J. Browne, "CAD/CAM Principles," Pearson Education, 2nd Edition, 1999.
3. Geometric Modeling, Michael E. Mortenson, Tata McGraw Hill, 2013.

Reference Books:

4. W. M. Neumann and R.F. Sproul, "Principles of Computer Graphics," McGraw Hill, 1989.
5. D. Hearn and M.P. Baker, "Computer Graphics," Prentice Hall Inc., 1992.

Online Resources:

1. NPTEL Lecture Series:

• https://nptel.ac.in/courses/112/102/112102101/

• https://nptel.ac.in/courses/112/104/112104031/

2. MIT OCW:

https://ocw.mit.edu/courses/mechanical-engineering/2-158j-computational-geometry-spring-2003/

Paper IV / Subject Name: Tribology		Code: MEE022C604
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objectives:

The objective of this course is to provide students with a fundamental understanding of the science of friction, wear, and lubrication, and their significance in the design, operation, and maintenance of mechanical systems. The course aims to develop the ability to analyze tribological interactions between contacting surfaces, select appropriate lubricants and materials, and apply tribological principles to improve the efficiency, reliability, and service life of engineering components

Course Outcomes

SI No	Course Outcome
CO 1	The students will be able to apply theories of friction and wear to various practical situations by analysing the physics of the process
CO 2	They will understand the various surface measurement techniques and effect of surface texture on Tribological behavior of a surface
CO 3	They will be able to select materials and lubricants to suggest a tribological solution to a particular situation
CO 4	The students will be able to design a hydrodynamic bearing using various bearing charts

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
Module I: Fundamentals of Tribology	**Friction and Wear Mechanisms**	Introduction to Tribology, significance and applications; theories of friction; genesis of friction; friction control techniques; surface texture and roughness measurement; stick-slip motion and frictional instabilities; introduction to wear; types of wear (adhesive, abrasive, corrosive, erosive, fatigue, fretting); theories of wear; wear prevention and surface engineering techniques.	15
Module II: Bearing Materials and Lubrication Principles	**Lubricants and Hydrodynamic Lubrication**	Tribological properties of bearing materials; classification and properties of lubricants; lubricant additives; lubrication regimes; Reynolds equation and its assumptions and limitations; hydrodynamic lubrication theory; idealized bearings; infinitely long plane pivoted and fixed shoe sliders; infinitely long and infinitely short journal bearings; Petroff's equation for lightly loaded journal bearings.	15

Module III: Hydrodynamic Bearing Design	**Journal Bearing Analysis and Design**	Finite journal bearings; design principles of hydrodynamic journal bearings; performance characteristics of journal bearings; hydrostatic lubrication; squeeze film lubrication; lubrication of circular and rectangular flat plates; lubrication under variable and alternating loads; piston pin lubrication; engineering applications of hydrodynamic and hydrostatic bearings.	15
Module IV: Advanced Tribology and Engineering Applications	**Elasto-hydrodynamic and Specialized Bearings**	Elasto-hydrodynamic lubrication (EHL); pressure-viscosity effects in Reynolds equation; Hertz contact theory; Ertel-Grubin equation; lubrication of spheres, gear teeth, and rolling element bearings; air-lubricated bearings; tilting pad bearings; recent advances and industrial applications of tribology in automotive, aerospace, manufacturing, and energy systems.	15
TOTAL			60

References:

1. Cameron, "Basic Lubrication Theory", Ellis Horwood Ltd, 1981.
2. Principles in Tribology, Edited by J. Halling, 1975
3. Fundamentals of Fluid Film Lubrication – B. J. Hamrock, McGraw Hill International,1994 Model Curriculum of Engineering & Technology PG Courses [Volume -II][173]
4. D.D. Fuller, "Theory and Practice of Lubrication for Engineers", John Wiley and Sons,1984.
5. "Fundamentals of Friction and wear of Materials" American Society of Metals.
6. Introduction to Tribology of Bearings –B. C. Majumdar, A. H. Wheeler &co. pvt. Ltd 1985.
7. T.A. Stolarski, "Tribology in Machine Design".

Course: B.Tech. (M.E)	SYLLABUS	Semester: VI
Paper V (Elective II) / Subject Name: Hydraulic Machines		Subject Code: MEE022D6061
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: T

Course Objective:

- To introduce students to turbines, pumps etc
- To enable the students to know the operation of fluid machines.
- To provide students thorough understanding of velocity triangles, thermodynamic plots and losses in turbo-machinery
- To introduce students to turbines, pumps etc

Detailed Syllabus:

Modules	Topics/ Course content	Hours
I.	**Hydraulic Turbines** Introduction: Classification of Fluid Machines, Impulse momentum equation, Impact of jet: Introduction to hydrodynamic thrust of jet on a fixed and moving surface (flat & curve), effect of inclination of jet with the surface. Hydraulic Turbines: Classification of turbines, Impulse turbines, constructional details, velocity triangles, power and efficiency calculations, governing of Pelton wheel.	15
II.	**Reaction Turbines:** Francis and Kaplan turbines, constructional details, velocity triangles, power and efficiency calculations, draft tube, cavitation in turbines, principles of similarity, unit and specific speed, performance characteristics, selection of water turbines	15
III.	**Centrifugal Pumps**: Classifications of centrifugal pumps, vector diagram, work done by impellor, efficiencies of centrifugal pumps, specific speed, cavitation, performance characteristics. **Reciprocating pump:** Positive Displacement Pumps: Reciprocating pump theory, slip and coefficient of discharges, indicator diagram, effect of acceleration, work saved by fitting air vessels, performance	15
IV.	**Fans and Blowers** Centrifugal Fans - Blowers Details - Inducers-Backward and Radial Blades. Axial flow fans. Fan laws **Fluid systems:** Hydraulic accumulator, Intensifier, Hydraulic press, theory of hydraulic coupling and torque converters, Water Lifting Devices: Hydraulic ram, Jet pumps, Airlift pumps, Gear and Vane pumps.	15
Total		**60**

Text Books:

1. C.S.P. Ojha, R. Berndtsson, "Fluid Mechanics and Machinery", Oxford University Press 4th edition 2012
2. R.K. Bansal, “Fluid Mechanics and Hydraulic Machines”, Laxmi Publications; Tenth edition, 2018

Reference Books:

1. Yahya.S.M., "Turbine, Fan and Compressors",TMH,2002.

Course: B.Tech. (M.E)	**SYLLABUS**	**Semester: VI**
Paper VI (Elective I) / Subject Name: Gas Dynamics and Jet Propulsion		
		Subject Code: MEE022D6031
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Detailed Syllabus:

Modules	Topics / Course content	Hours
I.	**Thermodynamic analysis cycles:** Simple Gas Turbine and Practical Gas Turbine cycles – Heat Exchanger cycle – Reheat cycle – Intercooled cycle etc. –- Intercooled cycle with Heat Exchanger - Intercooled cycle with Heat Exchanger and Reheat. **Combustion Systems:** Parts of Combustion Systems - Factors effecting Combustion Chamber Design – Mixing and Dilution – Types of Combustion Chambers.	**15**
II.	**Centrifugal Compressors:** Essential Parts – Idea Energy Transfer – Velocity Tringles – Blade Shapes and their analysis – Diffuser – Volute Casing – Losses – Surging and Choking – multi spool axial compressor and turbofan **Axial Flow Compressors:** Geometry and Working Principle – Stage velocity – Compressor Stage Efficiency – Losses – Degree of Reaction – Performance Characteristics – Stalling, surging and choking	**15**
III.	**Axial and Radial flow turbines** Introduction - Turbine stage – Velocity Triangle - Work Done -Degree of Reaction - Losses – Efficiency - Multi-staging of Turbine - Turbine cooling methods - Turbine Cooling Technology - Radial Turbine Aerodynamics and Thermodynamics.	**15**
IV.	**Jet Propulsion and Rocket Propulsion:** Ramjet Engine – Pulse Jet Engine – Turboprop Engine – Turbojet Engine – Thrust Equations and its applications – Efficiencies – Classifications of Rockets – Principle of Rocket Propulsion – Optimum Expression Ratio for Rocket. **Environmental Considerations:** Air Pollution – N0x Formation and Reduction - Noise and Noise Reduction.	**15**
Total		**60**

Text Books:

1. V Ganesan, “Gas Turbine”, McGraw – Hill Education Private Limited, 3rd Edition, 2017
2. Zucrow-Wiley, “Aircraft & Missile propulsion”,New York-1958

Reference Books:

1. A. H. Shapiro, “Compressible fluid flow”, The Ronald Press, New York-2002
2. S. M.Yahya, “Fundamentals of compressible flow with aircraft and rocket propulsion”, New Age International (P) Ltd.-2007

3. Liepman & Roshko, “Elements of gas dynamics”, Wiley, NewYork-1957
4. M.J. Zucrow & Joe D.Holfman, “Gas dynamics”, Krieger Pub. Co.-1985

DETAILED SYLLABUS OF 7th SEMESTER

Paper I / Subject Name: Finite Element Method		Code: MEE022C701
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objectives

- To provide understanding of the general steps of finite element method.
- To introduce the basic finite element formulation techniques.
- To derive the equations for 1D and 2D problems.
- To formulate and solve basic problems in heat transfer, solid mechanics and fluid mechanics.
- To derive the shape functions in terms of natural coordinates
- To solve any problems by any FE software such as ANSYS or be able to develop a code by their own.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the basic concepts of stress, strain, and governing differential equations used in FEM	**BT 1**
CO 2	Explain the formulation of shape functions and element matrices for 1D and 2D problems.	**BT 2**
CO 3	Apply the finite element procedure to solve simple structural, thermal, or fluid problems.	**BT 3**
CO 4	Analyze the results of FEM simulations and evaluate the influence of mesh quality and boundary conditions.	**BT 4**

Detailed Syllabus:

Modules	Topics / Course content	Hours
I.	**Finite Element Discretization,Direct stiffness methods and FE approximation methods** Basic concept of structural modelling, Introduction to FE modelling, Solving Axial rod and bending of beams by direct stiffness method – Pre-processing, solving, Post processing. Extensions to plane truss problems. Variational form – principle potential energy, Galerkin, RITZ, Least square method, collocation method	**15**
II	**1D & 2D Finite element formulation, Numerical Integration** Finite element formulation-completeness, compatibility, shape functions for c^0 and c^1 element, RITZ method, Galerkin method for axial rod problem, Steady state heat conduction problem, heat conduction in polar coordinates and fluid problems. Introduction to types of 2D element, triangular element, tetrahedral element, brick element and rectangular element, pascal's triangle, CST element- 3 node , 6 node and 10 node. Fluid flow problem. Introduction to quadrature, One Dimensional Integration Formulae - Gauss Quadrature 2 point and 3 point.	**15**

III.	**2D FEM : Detailed Discussion** Introduction- Natural Coordinates and Iso-Parametric, Sub-Parametric and Super-Parametric Elements, Four-Noded Quadrilateral Elements, Serendipity Elements, Eight-Noded Curvilinear Elements.	**15**
IV.	**Finite Element Formulations:** Finite element formulation for plane stress and plane strain problems, vibrations of a rod, vibrations of a beam, Thin plate formulation, Thick-plate formulation. Time history problems. Solving FEM Problems on a Computer: FEM Package – ANSYS, MATLAB.	**15**
Total		**60**

Text Books:

1. U.S Dixit, "Finite Element Methods for Engineers". Cengage Learning; 1st edition, 2009
2. R.D. Cook, D.S. Malkus and M.E. Plesha, "Concepts and Applications of Finite Element Analysis", Wiley; Fourth edition (2007)

Reference Books:

1. T.R. Chandrupatla and A.D. Belegundu, "Introduction to Finite Elements in Engineering:, Prentice Hall of India.
2. J.N. Reddy, "An Introduction to the Finite Element Method", McGraw-Hill.
3. S. S. Rao, "Finite Element Analysis", Elsevier Butterworth-Heinemann

Paper II / Subject Name: Renewable Energy Engineering		Code: MEE022D702
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objectives:

- To acquire knowledge of technical competency combined with research to generate innovative solutions in Energy engineering.
- To be acquainted with a variety of options in energy sources.
- To prepare the students to exhibit a high level of professionalism, integrity, environmental and social responsibility, and life-long independent learning ability with environment in mind.

Course Outcome:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and explain fundamental concepts and terminologies in energy engineering and energy management	**BT 1**
CO 2	Describe different energy systems, energy audit practices, and relevant management strategies.	**BT 2**
CO 3	Apply scientific and engineering principles to evaluate the performance of various energy systems.	**BT 3**
CO 4	Analyze energy consumption patterns and propose management solutions for improving energy efficiency in systems.	**BT 4**

Detailed Syllabus :

Modules	Topics	Course content	Hours
I.	**Introduction**: **Solar Energy**	Basic concepts of energy; Introduction to Renewable Energy Technologies; Energy and Environment – global warming, acid rains, depletion of ozone layer; Global and Indian Scenario of renewable energy sources; Energy storage - necessity and energy storage methods Fundamentals; Solar Radiation; Estimation of solar radiation on horizontal and inclined surfaces; Measurement of solar radiation data Basics of thermodynamics and heat transfer; Flat plate collector; Evacuated Tubular Collector; Solar air collector; Solar concentrator; Solar distillation; Solar cooker; Solar refrigeration and air conditioning; Thermal energy storage systems. Solar cell Fundamentals; Characteristics and classification; Solar cell: Module, panel and Array construction; Photovoltaic thermal systems	**20**

II.	**Wind Energy**	Introduction; Origin and nature of winds; Wind turbine siting; Basics of fluid mechanics; Wind turbine aerodynamics; wind turbine types and their construction; Wind energy conversion systems. **Fuel cells**: Overview; Classification of fuel cells; Operating principles; Fuel cell thermodynamics	**20**
III.	**Biomass Energy**	Introduction; Photosynthesis Process; Biofuels; Biomass Resources; Biomass conversion technologies; Urban waste to energy conversion; Biomass gasification.	**10**
IV.	**Other forms of Energy**:	Introduction: Nuclear, ocean and geothermal energy applications; Origin and their types; Working principles.	**10**
Total			**60**

Text /Reference Books:

1. O.P. Gupta, "Energy Technology", Khanna Book Publishing, New Delhi.
2. V.V.N. Kishore, "Renewable Energy Engineering and Technology: Principles and Practice," Routledge, 1st Edition, 2019.
3. N. Jenkins and J. Ekanayake, "Renewable Energy Engineering," Cambridge University Press, 1st Edition, 2017.
4. G. Boyle, "Renewable Energy," OUP Oxford, 2nd Edition, 2009

Online Resources:

1. https://onlinecourses.nptel.ac.in/noc22_ph44/preview
2. https://onlinecourses.swayam2.ac.in/nou22_ge71/preview

Paper III / Subject Name: Industrial Engineering		Code: MEE022O703
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objective:

1. To provide knowledge on machines and related tools for manufacturing various components.
2. To understand the relationship between process and system in manufacturing domain.
3. To identify the techniques for the quality assurance of the products and the optimality of the process in terms of resources and time management

Course Outcomes:

On successful completion of the course, the students will be able to:		
Sl. No.	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Recall the fundamental principles of production and operations management	**Remember (BT1)**
CO2	Explain the relationship between market demand and production capacity	**Understand (BT2)**
CO3	Apply resource planning techniques to optimize production processes and operations	**Apply (BT3)**
CO4	Analyze the impact of quality, cost, and time management tools on market competitiveness	**Analyze (BT4)**

Detailed Syllabus:

Modules	Topic	Course Content	Periods
I	Project Management & Forecasting	Scope of production management. Production system and resources (machines, tooling, etc.); Types of production (batch, flow and unit), Roles of line supervisors and production managers. Project life cycle: concept phase, Project planning, Forecasting, forecasting methods, moving average, exponential smoothing, Delphi technique, market survey analysis	15
II	Production Planning & Control	Facility planning, plant layout: process, product, group and fixed, motion study, time study, method study Project Monitoring and control: Project networks: CPM and PERT, critical path, re-scheduling, Project Crashing	15
III	Planning and Inventory Control	Aggregate production planning, materials requirement planning; Sequencing, line balancing, Gantt charts, Inventory Management: Economic order Quantity, discount models, inventory models, practical inventory control models, JIT; Supply chain and management, Modern manufacturing	15

IV	Operation Management	Linear programming, objective function and constraints, graphical method, Simplex and duplex algorithms, transportation assignment; Simple queuing theory models; Traveling Salesman problem; Network models: shortest route, minimal spanning tree, maximum flow model.	15
Total			60

Textbooks and References:

1. E.O Doebelin and Dhanesh Manik, “Measurement Systems”, McGraw Hill, 2017

2. Bewoor & Kulkarni, “Metrology & Measurement” Tata McGraw Hill, 2009.

3. D. James, and S, Meadow, “Geometric Dimensioning and Tolerancing”, Marcel Dekker, 1995

4. Madhav S. Phadke, Quality Engineering using Robust Design, Prentice Hall, 1989

Paper IV / Subject Name: Internship Evaluation (Seminar) **Subject Code: MEE022C706**

L-T-P-C – 0-0-0-0 **Credit Units: 01** **Scheme of Evaluation: P**

Objective

- To acquire industrial experience.

Course Outcomes:

On successful completion the student will be able to:

- Acquire and apply fundamental principles of engineering.
- Update with all the latest changes in technological world.
- Identify, formulate and model problems and find engineering solution based on a systems approach

The students need undertake minimum 3 weeks summer training in research institute/organization or industries at the end of the 6th semester. A report needs to be submitted in the start of 7th semester. The students will have to give a presentation on the summer training undertaken during the 6th semester.

Paper VI / Subject Name: Internship Evaluation		**Subject Code: MEE022C706**
L-T-P-C – 0-0-0-5	**Credit Units: 05**	**Scheme of Evaluation: P**

Objective

- To acquire industrial experience.

Course Outcomes:

On successful completion the student will be able to:

- Acquire and apply fundamental principles of engineering.
- Update with all the latest changes in technological world.
- Identify, formulate and model problems and find engineering solution based on a systems approach

The students need undertake minimum 3 weeks summer training in research institute/organization or industries at the end of the 6th semester. A report needs to be submitted in the start of 7th semester. The students will have to give a presentation on the summer training undertaken during the 6th semester.

Paper VII / Subject Name: Engineering Project-1 (Design & Analysis)		Code: MEE022C717
L-T-P-C –0-0-0-5	Credit Units: 5	Scheme of Evaluation: P

Objectives:

- This course is aimed to provide more weightage for project work.
- The project work could be done in the form of a summer project or internship in the industry or even a minor practical project in the college.
- Participation in any technical event/ competition to fabricate and demonstrate an innovative machine or product could be encouraged under this course.

DETAILED SYLLABUS OF 8th SEMESTER

Paper II/ Subject Name: Refrigeration and Air Conditioning		Subject Code: MEE022C802
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: T

Objectives:

1. To familiarize with the terminology associated with refrigeration systems and air conditioning 2. To understand basic refrigeration processes
2. To understand the basics of psychrometry and practice of applied psychrometrics
3. To acquire the skills required to model, analyse and design different refrigeration as well as air conditioning processes and components

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the fundamental principles, terminologies, and components involved in refrigeration and air conditioning systems.	**BT1**
CO 2	Explain the working of various refrigeration cycles (e.g., vapor compression, absorption) and psychrometric processes.	**BT2**
CO 3	Apply thermodynamic principles to design and analyze basic refrigeration and air conditioning systems.	**BT3**
CO 4	Analyze system performance using COP, refrigeration effect, and psychrometric charts under varying operating conditions.	**BT4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Introduction To Refrigeration And Air-Conditioning**	Basic Definitions of Refrigeration and Air-Conditioning; History of Refrigeration; Natural and Artificial Refrigeration Methods; Techniques to produce low temperatures; Applications of Refrigeration; Refrigerants- Classification, Nomenclature, Desirable Properties, Selection. Air Refrigeration Cycles - reversed Carnot cycle; Bell-Coleman cycle analysis; various methods of Aircraft Refrigeration: Analysis, Merits and demerits.	**10**
II.	**Vapor Compression**	Ideal VCR cycle (Working, Analysis and Limitations); Standard VCRS (Working and Analysis); Methods to improve	**10**

	Refrigeration System	performance of VCR; Multi-Stage VCRS; Cascade Refrigeration. **Components of Refrigeration Systems:** Compressors: Positive Displacement (Reciprocating and Rotary); Dynamic (Centrifugal and Axial) Compressors; Condensers and Evaporators (Both Natural and Forced Convection type); Expansion Devices and other components of the system	
III.	**Vapor Absorption Systems** **Psychrometry:**	Working and Analysis; Absorbent - Refrigerant combinations; Water-Ammonia Systems; Water-Lithium Bromide System; Modified Version of Aqua-Ammonia System with Rectifier and Analyzer Assembly. Brief Discussion on (i) Steam-Jet refrigeration system; (ii) Vortex tube refrigeration; (iii) Thermoelectric refrigeration system; and (iv) Magnetic refrigeration **Psychrometry:** Classification of Air-Conditioning Systems; ASHRAE Nomenclature; Applications of Air-Conditioning; Psychrometry - Air-water vapor mixtures; Psychrometric Properties; Psychrometric or Air-Conditioning processes; Psychrometric Chart.	**10**
IV.	**Air-Conditioning Systems**	Classification of Air-Conditioning Systems; Psychrometry of Air-Conditioning Systems; Thermal Comfort (Definition and Psychrometric Properties for Thermal Comfort); Mathematical Analysis of Air-Conditioning Systems; Cooling and Heating Load Estimation; a brief discussion on Ventilation.	**10**
TOTAL			**40**

Text Books:

1. Singh, Sadhu., Refrigeration and Air conditioning, Khanna Book Publishing Edition, 2021.
2. Domkundwar, Refrigeration and Air conditioning, Dhanpat Rai.
3. Gosney, W.B, Principles of Refrigeration, Cambridge University Press, 1982.

Reference Books:

1. Stoecker, W.F. and Jones, J.W., Refrigeration and Air conditioning, Tata McGraw Hill, 1986.
2. Arora, C.P., Refrigeration and Air conditioning, Tata McGraw Hill, 2nd Edition, 2000.
3. Kuehn, T.H., Ramsey, J.W. and Threlkeld, J.L., Thermal Environmental Engineering, 3rd Edition, Prentice Hall, 1998.

Online Resources:

https://onlinecourses.nptel.ac.in/noc22_me135/preview

Paper II/ PE 3		**Subject Code: MEE022D8021**
Subject Name: Computational Fluid Dynamics (CFD)		
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objectives:

- To provide students with the ability to simulate fluid flow, heat, and mass transfer phenomena in a wide range of engineering and natural systems.
- To highlight the necessity of accurate flow prediction for the design and optimization of thermal and fluid-based devices.
- To develop an understanding of the fundamental principles and numerical methods of CFD, with an emphasis on practical applications and design optimization in modern engineering systems.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and explain the fundamentals of aerodynamic flow and the mathematical form of Navier–Stokes equations.	**BT 1**
CO 2	Describe and set up a CFD problem with appropriate geometry, meshing, and boundary conditions using commercial tools,	**BT 2**
CO 3	Apply CFD software to solve simplified flow problems and simulate engineering systems with appropriate physical parameters	**BT3**
CO 4	Analyze CFD results, validate them against available data, and interpret the findings in the context of engineering applications.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Topics / Course content**	**Hours**
I.	**The Equations of Fluid Dynamics**	General form of a Conservation law: equation of mass conservation, conservation law of momentum, conservation equation of energy. The Navier-Stoke's equation, Reynold's Averaged NS equation, Boundary Layer Approximations, The distributed loss model, The inviscid flow model, Euler equations, steady inviscid rotational flow, The potential flow model. Non-dimensionalization of Continuity, NS and Energy equations and its applications	**15**

II.	**Basic discretization techniques:**	Types of Partial Differential Equations and its applications, Truncation Error analysis, Consistency, Stability, Convergence. The finite difference method: Conversion of PDE to FDE using Taylors Series and 3-Point Method. The finite volume method and conservative discretization: Integral Method - Meshing and Grid Generation	**15**
III.	**Solutions of Poisson's and Laplace equation**	FTCS method- FTFS method etc. Solution of 1D Heat conduction equation – Simple explicit method – ADI method – DuFort-Frankel method – Crank Nicolson method etc.	**15**
IV.	**Solution of Berger's equation for Inviscid and Viscous flow using implicit and explicit methods**	Solution of Berger's equation for Inviscid and Viscous flow using implicit and explicit methods. Solution of Wave equation – Euler explicit method – Lax method – Upwind method – MacCormack method – Leap Frog method etc	**15**
Total			**60**

Text Books:

1. Johan D Anderson, JR., "Computational Fluid Dynamics" McGraw-Hill Education Private Ltd, New Delhi.

Reference Books:

1. Johan D Anderson, JR., "Computational Mechanics and Heat Transfer" Taylor & Francis.
2. T. J. Chung, "Computational Fluid Dynamics" Cambridge University Press.
3. Verlag., "Computational techniques for Fluid Dynamics" Fletcher and Springer.
4. Charlse and Hirch., "Numerical Computation of Internal and External flows" John-Wiley
5. John C. Tannehill, Dale A. Anderson and Richard H. Pletcher, "Computational Fluid Mechanics and Heat Transfer" Taylor &Francis.

Paper IV / Subject Name: Tool Engineering		Code: MEE022D8031
L-T-P-C –4-0-0-4	Credit Units: 4	Scheme of Evaluation: T

Objectives:

1. Understand the fundamentals of tool engineering, process planning, and the selection of cutting tools used in manufacturing industries.
2. Develop knowledge of locating, clamping, jigs, and fixtures for improving machining accuracy, productivity, and quality.
3. Apply the principles of press tool design, sheet metal operations, and die selection for manufacturing applications.
4. Familiarize students with various dies and moulds used in metal forming and plastic processing, along with their industrial applications.

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain the fundamentals of tool engineering, process planning, and the selection of cutting tools and tool materials for manufacturing applications.**	**BT 2**
CO 2	**Apply the principles of location and clamping to select suitable locators, clamps, jigs, and fixtures for machining operations.**	**BT 3**
CO 3	**Analyze press working operations and perform basic calculations for the design and selection of press tools and cutting dies**	**BT4**
CO 4	**Select appropriate dies and moulds for bending, drawing, forging, extrusion, and moulding operations based on manufacturing requirements.=**	**BT 3**

Detailed Syllabus

Modules	Topics	Topics / Course content	Hours
I.	**Introduction to Tool Engineering and Cutting Tools**	Introduction to tool engineering: concept, objectives, scope and importance Tool engineering in manufacturing industries. Process planning and its role in productivity and quality. Universal acts and elements of manufacturing operations. Cutting tool materials: classification, properties and applications. Carbide inserts: types, applications and basic ISO designation. Tool holders and tool mounting systems for conventional	**15**

		machine tools. Overview of cutting tool re-sharpening methods	
II.	**Locating, Clamping Devices, Jigs and Fixtures**	Concepts of location and clamping. Degrees of freedom and 3-2-1 principle. Types and applications of locating devices. Types and applications of clamping devices. Introduction to jigs and fixtures Types, construction and applications of jigs and fixtures. Basic design considerations for jigs and fixtures.	**15**
III.	**Press Tools and Sheet Metal Dies**	Introduction to press working operations. Types and construction of press tools. Press tool components and working. Press tonnage and shut height. Die clearance and cutting force. Shear angle and its significance. Scrap strip layout and stock utilization. Stock stops, pilots, strippers and knockouts. Types and applications of cutting dies. Introduction to progressive dies	**15**
IV.	**Dies and Moulds**	Bending operations: bend allowance, bend radius and springback. Bending dies: types and applications. Drawing dies and blank size determination. Overview of drawing operations. Forging dies: types and applications. Introduction to extrusion dies. Plastic injection moulds. Blow moulds.	**15**
Total			**60**

Textbooks:

1. Fundamentals of tool design, ASTME, PHI.
2. Tool design. Donaldson & Lecain, TME.
3. Tool engineering, Doyal.
4. Principles of tool & jig design, M. H. A. Kempster

Paper IV / Subject Name: Engineering Project-2		Code: MEE022C804
L-T-P-C –0-0-0-8	Credit Units: 8	Scheme of Evaluation: P

Objectives:

- It is intended to start the project work early in the seventh semester and carry out both design and fabrication of a mechanical device whose working can be demonstrated. The design is expected to be completed in the seventh semester and the fabrication and demonstration will be carried out in the eighth semester.
- During VIIIth semester, each student is required to complete the project as per the plan made in the preliminary report submitted during the VIIth semester. At the middle of the VIIIth semester an Interim Evaluation will be carried out by the evaluation committee constituted in the previous semester. At the end of the semester, each student should also appear for Final Evaluation.
- Interim Evaluation of the project should be done at the middle of the eighth semester by the committee. Each student should submit a copy of the Interim Report of the Project before the committee. Also, copies of the Approval of Project and Preliminary Report shall be submitted to the evaluation committee. The committee will award marks based on the student-wise performance. The respective guide will award the individual internal marks